C H A P T E R 1

COURSE MANAGEMENT DETAILS

G E N E R A L

Aim

1. The aim of this course is to qualify the MARS 71B officer as a Clearance Diving Officer sub-occupation 71D.

REQUIREMENTS

2. Course Duration The duration of this course is 270 training days, broken down as follows

PHASE	LOCATION	TRAINING DAYS
01 WELDING	CFNES	5
02 DIVE MEDICINE	DCIEM TORONTO	15
03 BASIC EOD	CFB BORDEN	27
04 INTERNATIONAL EOD	EGLIN AFB	73
05 COMCMV	EGUERMINE BELGIUM	25
06 MAIN COURSE	FDU(A)	145
TOTAL TRAINING DAYS		270

3. Resources; Instructor allocation and location requirements are as follows;

LOCATION	TRAINING	INSTRUCTOR	HOURS
FDU		Officer - 71d b.3 Instructor-PO1 342 HB standby - MS/LS 342	
DCIEM	Diving medicine and theory	PO1 342 ADMO (ADXB qual)	
CFB Borden	Basic EOD	WO 921 PO2 342	
CFFS Halifax	Welding and cutting	PO1 HT 321	
Eguermine	COMCMV	Out service training	
Eglin AFB	International EOD	Out service training	

4. Course Capacity;

a. maximum - 8
b. optimum - 6
c. minimum - 4

5. Other requirements None

MANAGING AUTHORITY

6. The managing authority is CMS/DNPP. Courses will be managed using Individual Training Management Information System (ITMIS). It involves nominating individuals, course scheduling, matching personnel and serials for training, costing resources for training, course loading, course reporting and tracking training results.

COURSE PRE - REQUISITIES.

7. Candidates for this course shall:

a. be a volunteer;
b. be physically fit IAW CFAO 50-1 and CFAO 43-2 Annex A;
c. recommended by the Commanding Officer;
d. recommended by PSO
e. meet military swimming standard IAW STD (AHCU);
f. be a current, qualified STD officer (AEJF)
g. be sub-occupation qualified MARS 71B or 71C
h. successfully complete a two week preliminary evaluation at a FDU
i. be obligated to sign a four (4) year return of service contract to commence upon completion of CLDO training: and
j. selected by NDHQ; and
.

RELATED DOCUMENTS.

8. A-PD-150-002/PP-005, CF Manual of Officer's occupational structure, MARS 71 section 3, duty area C - diving ops

EVALUATION.

9. Candidates must successfully complete all performance Objectives as documented in the QSP. Assessment details are contained in Chapter 3 of this document.

QUALIFICATION.

10. Qualification is as follows:

a. In order to attain 71D qualification (ADFR) the following four phases must be completed

(1) phase one (AEXQ).
(2) phase two
(3) phase three (00991)
(4) phase four (AEDV)

b. Related Specialties The following occupation specialty specifications dictate the requirement for additional training related to the CLDO sub-occupation

(1) 00992 mine warfare planning for international officers (Ingleside Tx.)
(2) 00991 phase two - out service training (Eguermine)
(3) Not assigned - staff officer (6 mos Eguermine)
(4) AEDW - Salvage diving officer;
(5) Experimental diving facility operator (level 1) (EDU); and
(6) Experimental diving facility operator (level 2) (EDU)

SPECIAL INSTRUCTIONS.

11. Due to the PT required for CLDO's, PT should normally be conducted three (3) times per week and shall not be considered as TAT (MARCORD 9-30)

12. TAT shall be granted IAW MARCORD 9-30)

13. While on course the trainee is entitled to Casual Clearance diving allowance IAW QRO 205.345.

14. Qual AEXQ will be achieved through instruction at CFSEME Borden.

15. Qual 00991 will be achieved through instruction at COMCMV

16. Qual AEDV will be achieved through instruction at NAVSCOLEOD, Eglin AFB, LO international EOD

C H A P T E R 2

PERFORMANCE OBJECTIVES

PO 401

1. Performance: Supervise RCC Operations

2. Conditions:

   a. Given:

      (1) qualified RCC operating team
      (2) RCC;

         (a) DUOCOM
         (b) Main FDU
         (c) SUBSAR
         (d) YDT; and
         (e) containerized RCC system.

      (3) checklist
      (4) references; and
      (5) dive task

   b. Denied: supervisory assistance;

   c. Environmental: N/A.

3. Standard: The Diving Officer shall, IAW applicable references:

   a. operate multi-lock and DUOCOM RCCs
   b. supervise RCC operations including;

      (1) state RCC rules and regulations
      (2) state RCC SOPs;
      (3) state the general characteristics of RCCs
      (4) produce a RCC system schematic drawing;
      (5) pre-dive equipment checklist is correct;
      (6) decompression procedures;
      (7) safe conduct of the dive;
      (8) control the dive team throughout the dive;
      (9) correct response to RCC emergencies;
      (10) explain DUOCOM RCC
      (11) completeness and accuracy of reports and returns

4. References: A2,A5,A9,A17,A20, A23 to A31, A41, C6,C7.

5. Specification Task Number: CT002

6. Supporting Skills: CS004,

7. Supporting Knowledge: CK001,CK005,CK013 to CK015

8. Training Limitations: None Identified

PO 402

1. Performance. Supervise Emergency Treatment of Diving Ailments.

2. Conditions;

   a. Given;

      (1) diving teams;
      (2) RCC;
      (3) diving emergency scenario; and
      (4) ancillary medical equipment.

   b. Denied; assistance

   c. Environmental N/A

3. Standard. The Diving Officer shall, IAW applicable refs, supervise treatment of diving ailments including;

   a. apply diving physics
   b. explain diving physiology and medicine;
   c. correctly diagnose ailments;
   d. explain hyperbaric treatments;
   e. explain diving ailments requiring treatments; and
   f. direct/ control emergency treatment and/or evacuation for definitive treatment.

4. References; A2 to A5, A23 to A26, A29, A33, A41, C1, C2,C6,C7

5. Specification Task Numbers. CT002, CT003, CT004

6. Supporting Skills. CS004, CS005, CS006

7. Supporting Knowledge. CK001, CK013 to CK015

8. Training Limitations; None identified.

PO 403

1. Performance Conduct Basic EOD procedures

2. Conditions.

   a. Given;

      (1) qualified EOD personnel
      (2) task;
      (3) explosives;
      (4) demolition kit;
      (5) ordnance;
      (6) references

   b. Denied; assistance

   c. Environmental within environmental limits prescribed by applicable CF manuals to include Environmental and Fisheries Acts.

3. Standard. The standard for this PO will be achieved IAW CFSEME QS EOD-BASIC (QUAL Code - 05925) including:

   a. examine ammunitions and explosives to determine their hazards;
   b. dispose of ammunition and explosives by electric and non-electric methods;
   c. conducts EOD/ IED assistant operator duties; and
   d. perform logistics and administrative duties.

4. References. A7, A10,A12,A39,A43,A44,B1

5. Specification Task Numbers; CT012

6. Supporting Skills; CS014

7. Supporting Knowledge; CK024

8. Training Limitations; This PO will be met through instruction at CFSEME, CFB Borden.

PO 404

1. Performance. Supervise SCBA dives.

2. Conditions:

   a. Given:

      (1) appropriate dive team equipped with:

         (a) CABA;
         (b) CCDA; and
         (c) CUMA.

      (2) references;
      (3) dive task; and
      (4) support equipment

   b. Denied: assistance

   c. Environmental:

      (1) within environmental limits prescribed by applicable CF manuals, including ice; and
      (2) day and night.

3. Standard. The Diving Officer shall, IAW applicable refs, conduct and supervise SCBA operations including:

   a. CABA;
   b. CCDA; and
   c. CUMA.

4. References. A2, A3, A9, A14, A28, A30, A31, A33, A47, A51, C6.

5. Specification Task Numbers. CT001 CT005, CT006, CT007, CT008, CT015.

6. Supporting Skills CS001, CS003, CS009, CS011, CS016.

7. Supporting Knowledge CK001 to CK005, CK008 to CK010.

8. Training Limitations. Dependant upon scheduling of other sections.

PO 405

1. Performance: Supervise demolition operations.

2. Conditions:

   a. Given:

      (1) task;
      (2) qualified EOD personnel;
      (3) explosives;
      (4) demolition kit;
      (4) targets;
      (5) support platform for underwater tasks;
      (6) support vehicles for surface task; and
      (7) surface and underwater demolition site;

   b. Denied: assistance

   c. Environmental within environmental limits prescribed by applicable CF manuals to include Environmental and Fisheries Acts.

3. Standard: The Diving officer shall, IAW applicable refs, plan, conduct and supervise surface and underwater demolition operations to include;

   a. adherence to rules and regulations;
   b. determine charge type, size and correct placement; and
   c. carryout demolition procedures.

4. References: A7, A10, A12, A39.

5. Specification Task Number: CT009, CT012.

6. Supporting Skills: CS015.

7. Supporting Knowledge: CK024, CK031.

8. Training Limitations: requires the use of CFB Gagetown range.

PO 406

1. Performance: Supervise SSBA operations

2. Conditions:

   a. Given:

      (1) dive team equipped with;

         (a) RCC; and
         (b) diving equipment

      (2) mixed gas and air diving system;
      (3) dive tasks;
      (4) checklists;
      (5) SOPs;
      (6) support equipment; and
      (7) support vessel
      (8) references;

   b. Denied: assistance;

   c. Environmental:

      (1) within environmental limits prescribed by CF manuals;
      (2) day and night

3. Standard: The Diving Officer shall, IAW applicable refs, conduct and supervise SSBA Air and $He0_2$ operations including:

   a. SL17b; and
   b. lightweight surface supplied diving apparatus.

4. References: A2, A3, A4, A9, A17, A19 to A22, A51, C6, C7

5. Specification Task Number: CT001, CT005, CT006, CT015.

6. Supporting Skills CS002, CS009, CS011,

7. Supporting Knowledge: CK001 to CK005, CK008, CK011

8. Training Limitations: Dependant on scheduling of other sections

PO 407

1. Performance. Supervise BDR operations

2. Conditions.

   a. Given;

      (1) dive team;
      (2) support personnel;
      (3) references;
      (4) checklists;
      (5) task; and
      (6) support vessel/ vehicles

   b. Denied. assistance

   c. Environmental.

      (1) day and night; and
      (2) within environmental limits prescribed by CF manuals, including ice

3. Standard. The diving Officer shall, IAW applicable refs, conduct and supervise BDR operations including;

   a. underwater inspections and repairs;
   b. ROV ops
   c. carry out u/w searches/ recovery operations
   d. light salvage ops; and
   e. completeness and accuracy of reports and returns

4. References. A1,A3,A4,A9,A18,A37,A46,A49,A50,A54,C6,C7

5. Specification Task Numbers. CT001, CT005, CT006, CT015

6. Supporting Skills. CS001, CS002, CS007, CS009, CS013

7. Supporting Knowledge. CK001 to CK005, CK007 to CK009, CK011, CK018, CK030

8. Training Limitations None Identified.

PO 408

1. Performance. Coordinate MCM operations

2. Conditions;

   a. Given

      (1) task
      (2) references; and
      (3) MCM vessels.

   b. Denied. assistance

   c. Environmental. N/A

3. Standard. The Diving officer shall, IAW applicable refs, plan, coordinate and execute MCM operations including;

   a. MW organization and threats;
   b. mine and mine-laying;
   c. mine sweeping;
   d. mine hunting;
   e. MCM general operations; and
   f. naval Control of Shipping.

4. References. A11,A38,A55,B4 to B7, B13,B15,B17,B18,B19

5. Specification Task Numbers. CT014

6. Supporting Skills. CS001,CS011,CS014,CS015.

7. Supporting Knowledge. CKOO6 to CK010, CK021,CK024,CK027 to CK029

8. Training Limitations The standard for this PO will be achieved through out-service training IAW EGUERMINE Ostende, Be, COMCMV (Qual code 00990).

PO 409

1. Performance. Supervise EOD operations

2. Conditions.

   a. Given;

      (1) task;
      (2) qualified EOD personnel;
      (3) explosives;
      (4) demolition kit;
      (5) EOD equipment;
      (6) references; and
      (7) support vessel/ vehicle

   b. Denied. assistance

   c. Environmental. within environmental limits prescribed by applicable refs including Environmental and Fisheries Acts.

3. Standard. The Diving Officer shall, IAW applicable refs, plan , conduct and supervise surface and underwater EOD operations including;

   a. detection of explosive ordnance;
   b. identification of explosive ordnance;
   c. evaluation of explosive ordnance;
   d. render safe/ neutralize explosive ordnance; and
   e. disposal of explosive ordnance.

4. References. A10, A12, A43, A44, B3, B8.

5. Specification Task Numbers. CT012, CT014.

6. Supporting Skills. CS001, CS015.

7. Supporting Knowledge. CK006, CK010, CK023, CK024, CK028.

8. Training Limitations. The standard for this PO will be achieved through out-service training at NAVSCOLEOD, Eglin AFB LO. international EOD (Qual code AEDV).

CHAPTER 3

ASSESSMENT OF TRAINEE

GENERAL

1. The trainees must pass all Performance Checks (PC's) to successfully pass the course.

2. The nature of diving operations demands that participants be totally committed to teamwork and co-operation. Candidates undergoing training for MARS 71D qualification will be therefore be expected to demonstrate consistently high standards of personal conduct during training.

3. Trainees with an unsatisfactory attitude towards teamwork and safety will be removed from training.

PROGRESS MONITORING

4. The progress of trainees is conducted on a continuous basis in the following manner

   a. at the end of each PO - through a personal interview
   b. PC/EC's ; and
   c. Oral/written counselling

5. The progress of trainees for module two (out-service training is conducted on a continuous basis IAW COMCMV training standards, EGEURMINE, Ostend, Be.

6. The progress of trainees for module three (out-service training is conducted on a continuous basis IAW international EOD training standards, NAVSCOLEOD, Eglin AFB.

7. Monitoring, in general, shall provide the trainee with

   a. early warning of trainee difficulty;
   b. assistance in order to continue training; and
   c. TRB guidelines as necessary

8. The instructor will ensure the following documentation is implemented for each trainee:

   a. trainee progress books/records to include EC/PC failures;
   b. master sheet to record all EC/PC results; and
   c. master sheet to record all dive time.

PERFORMANCE CHECKS

9. Performance Checks, based on the PC's stated in Chapter 2 of this QSP, will be the primary basis for a pass/fail assessment of all officers.

10. To pass the course, the officer must pass all PC's. The trainee is permitted one re-check per PO up to a maximum of three (3) re-checks per module throughout the course. POs are checked by a combination of PCs and ECs (practical and written). Detailed criteria for the marking of PO)s are contained in the MARS 71D QSP

11. POs 403, 408 and 409 will be met through instruction at the school listed in the Training limitations for each PO. The respective EOs are held in the training documentation at each school.

UNSATISFACTORY COURSE PROGRESS

12. When a trainee fails an EC/ PC, the trainee will be counselled by the course instructor prior to the recheck. Counselling will inform the trainee of the nature of the shortcomings, suggested corrective action and possible consequences of further failures.

13. Trainees must be given 24 hours between the initial check and the recheck to allow time for study.

14. Failure to pass any 1 EC/PC after 2 attempts – candidate will be referred to a Progress Review Board (PRB)

15. Failure to pass any 1 EC/PC after 3 attempts – candidate will be referred to a Training Review Board (TRB)

16. Failure to pass 2 different EC/PC's (first attempt) – candidate will be referred to a PRB

17. Failure to pass 3 different EC/PC's (first attempt) – candidate will be referred to a TRB

18. Course Removal for Other reasons

a. Trainees who request voluntary withdrawal shall be counselled, through the Divisional system, on the career implications of such actions. Every effort should be made to motivate the trainee to continue training. If the trainee persists in his request, the TRG officer shall have the trainee complete a "Request to Cease Training" memorandum which will be routed through the TRGO to the C.O. for approval. Copies of such requests shall be attached to each copy of the course report (CF 377)

b. Medical, compassionate or other such reasons may cause a student to miss portions of scheduled training. Should this occur, the TRGO shall review the particulars of each case and make a recommendation to the C.O. to either retain the student under training with remedial instruction or to cease training with a recommendation to attend a future course.

c. The nature of diving operations demands that participants be committed to teamwork and co-operation. Students who fail to demonstrate a consistently high standard in the qualities will be counselled with regards to their shortcomings and given appropriate time to improve their performance. If after counselling, there is no notable improvement in attitude, a TRB will be convened on the matter.

<u>COURSE REPORTS</u>

19. Form CF 377, Canadian Forces Report, shall be prepared and distributed IAW CFAO 26-1:

(1). Submit draft copies of CF 377 to Training chief;
(2) Proof read CF 377s and course certificates prior to trainees signing;
(3). Confirm CF 377's have been mailed.

20. Officer trainees passing this course will be graded with an A, B or C rating, computed as follows;

a. A - achieved all PO's without difficulty and demonstrated exceptional ability. This grade indicates that the trainees performance was superior to that of the majority of the trainees.

(1) In order to achieve an A grade, all practical PC/EC and written exams(EC) for each PO must be achieved on the first attempt.
(2) The trainee also demonstrated exceptional ability WRT enthusiasm, confidence and decisiveness
(3) Additionally, the trainee must achieve a minimum of 90% on the written exams

b. B - Achieved all PO's without difficulty.

(1) A recognized exception would be if a trainee experiences difficulty in one or more POs, but demonstrates exceptional ability in other POs sufficient to off-set these deficiencies, they may be awarded a B; and

c. C - one or more PO's failed on first attempt.

21. Trainees on this course will be numerically ranked (1, 2, 3…) according to;

a. PO/EO's achieved without difficulty; and

b. the number of PO/ EO's in which exceptional ability was demonstrated.

TRAINING REVIEW BOARD

22. All TRB requirements are contained in MARCORD 9-60 (VOL 1).

C H A P T E R 4

ENABLING OBJECTIVES

1. Performance: State RCC rules and regulations

2. Conditions:

   a. Given: references;

   b Denied. assistance

   c. Environmental. N/A

3. Standard: The officer shall, IAW references and SOPs:

   a. State RCC rules and regulations IAW refs including;

      (1) personnel responsibilities;
      (2) safety;
      (3) administrative and medical records; and
      (4) diving incidents and accidents.

4. Time: 3 x 50 minute periods:

5. Method/ Media: Lecture,

6. Reference: A3, A5

7. Training Aids:

   a. whiteboard;
   b. OHP;
   c. multi-media
   d. manuals

8. Test Details: Pass mark is 85 % on written exam

9. Remarks: None

10. Teaching Points:

    a. Explain personnel responsibilities and qualifications IAW ref A5 including;

       (1) CO (art 1202);
       (2) Diving Officer (art 1203 and fig 5-1-2 sht 1);
       (3) Chamber supervisor (art 1204 and fig 5-1-2 sht 1);
       (4) Diver (art 1205 and fig 5-1-2 sht 1)
       (5) Diving medical personnel (art 1206) including

          (a) advanced diving medical officer (ADMO);and
          (b) diving medical technician (fig 5-1-2 sht 1)

       (6) Chamber operator (fig 5-1-2 sht 2);
       (7) Time keeper (fig 5-1-2 sht 2);
       (8) Compressor operator (fig 5-1-2 sht 2);
       (9) Gas bank operator (fig 5-1-2 sht 2); and
       (10) Chamber attendant (fig 5-1-2 sht 2).

b. Explain safety regulations IAW ref A5 including;

(1) minimum pers required (art 1301);
(2) selection of hyperbaric treatment (art 1302 para 3-5);
(3) flying after diving (art 1303, para 2-3) review ref A3 art 130;
(4) hyperbaric chambers - other uses (art 1304);
(5) civil chambers (art 1305)
(6) equipment maintenance(art 1306).

c. explain administrative and medical records including;

(1) diving administrative orders (art 1402 and review CFAO 43-2)
(2) orders pertaining to diving (ref A3, annex 1A, page 1A-1); and
(3) technical orders and instructions (art 4113).

d. review diving incidents and accidents (ref A5, art 1501 and ref A3, annex A ,chap 1).

e. Explain equipment emergencies IAW ref A5 including;

(1) intro;
(2) chamber comms failure;
(3) compressor failure;
(4) power failure;
(5) O_2 failure
(6) fire suppression system failure;
(7) chamber fire;
(8) fire or contaminated atmosphere in chamber area;
(9) uncontrolled air loss; and
(10) uncontrolled air or gas entry.

EC 401.1

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERAL AND FACILITIES REQUIRED.

1. classroom

2. references

ARRANGEMENTS AND PROCEDURES

3. The officer will complete a written exam on the rules and regulations of RCC operation..

4. References are allowed, assistance is denied.

5. Pass-mark is 85 %.

6. Time required: 1 x 50 min periods

PART II - INSTRUCTIONS TO BE GIVEN TO THE TRAINEE

1. This EC will confirm your knowledge of RCC rules and regulations.

2. References are allowed; assistance is denied.

3. Pass-mark is 85 %.

4. Time allowed: 1 x 50 min periods

PART III ENABLING CHECKNOTES. Test plan;

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE
SAFETY		30
ADMINISTRATIVE		20
EQUIPMENT EMERGENCIES		20
PERSONNEL RESPONSIBILITIES		10
EXAM		20
TOTAL		

EO 401.2

1. Performance. State general characteristics of RCCs.

2. Conditions,

   a. Given;

      (1) RCC
      (2) home assignment (drawing)
      (3) training student; and
      (4) technical assistance.

   b. Denied;

      (1) assistance; and
      (2) references

   c. Environmental. N/A.

3. Standard The officer shall ;

   a. state the general characteristics of RCCs IAW ref A5, chap 4, sect 1 including;

      (1) fire suppression systems;
      (2) breathing gas systems;.
      (3) electrical systems;
      (4) associated hazards;
      (5) HP to LP air systems; and
      (6) uses of RCCs in CF.

   b. produce a schematic drawing of the following RCC systems IAW ref A5;

      (1) air supply and exhaust;
      (2) BIBS; and
      (3) fire suppression..

4. Time. 9 x 50 min periods.

5. Method/ Media.

   a. lecture - 2 periods
   b. work on drawing - 5 periods
   c. review - 1 period
   d. EC - 1 period.

6. References. A5 A25 A26

7. Training Aids;

   a. OHP and multi media systems;
   c. whiteboard; and
   d. applicable chamber including all associated systems

8. Test Details. Pass mark is 75 %
9. Remarks. nil

10. Teaching Points:

a. explain the general characteristics of hyperbaric chambers , including;

(1) CF use of hyperbaric chambers;
(2) chamber characteristics.;
(3) air compression and storage;
(4) air supply and exhaust;
(5) contamination from chamber materials emphasizing hyperbaric toxicity hazard;
(6) breathing gas storage and supply, review purity of compressed breathing air and gas;
(7) hyperbaric fire hazard; and
(8) chamber outer lock use.

b. produce a schematic, including;

(1) the following RCC systems;

(a) air supply and exhaust;
(b) BIBS; and
(c) fire suppression.

(2) show all valves, gauges, regulators and piping;

(3) use piping color code and symbology IAW ref A5, chap 4, sect 1, art 4112 and fig 4-12 to 4-14;

(4) submit completed drawings to instructor within two (2) weeks of tasking; and

(5) ensure neatness, legibility and completeness of drawings.

EC 401.2

PART I - ADMINISTRATIVE INSTRUCTIONS.

MATERIAL AND FACILITIES REQUIRED

1. Main RCC

2. Applicable CFTOs, SOPs and references.

3. Drawing materials

ARRANGEMENTS AND PROCEDURES

4. The officer will complete a written exam on the general characteristics of RCCs IAW classroom lecture.

5. Assistance and references are denied

6. Pass mark is 80 %

7. Time allowed 1 period

8. The officer will complete schematic drawing as per the standard (para 3b)

PART II – INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of the general characteristics of RCCs.

2. You are denied assistance and references.

3. Pass mark is 80 %

4. Time allowed 1 period.

PART III – ENABLING CHECK NOTES - TEST PLAN.

MAJOR CONTENT	SPECIFIC CONTENT	VALUE (%)
CHARACTERISTICS	CF Chambers	20
SYSTEMS	HP/LP air	15
	breathing gas	15
	electrical	15
	fire suppression	15
HAZARDS	fire	10
	toxicity	10

PART IV – CHECKLIST N/A

EO 401.3

1. Performance: Operate main RCC

2. Conditions:

a. Given:

(1). main RCC/SUBSAR/ portable RCC and support systems;
(2). support team;
(3). dive profile;
(4). references and SOPs;
(5). logs and records;
(6). applicable pre-post dive checklists; and
(7) supervision.

b. Denied. assistance.

c. Environmental. N/A

3. Standard: The officer shall,

a. state RCC SOPs IAW ref A5 chap 5;

b. operate the main RCC including pre-post dive routines IAW applicable RCC SOPs and checklists;

c. follow the dive profile and maintain:

(a) required descent/ascent rates to within +/- 2 fsw/min;
(b) communications;
(c) stop depths to within +/- 1 fsw; and
(d) chamber ventilation IAW ref A5, art 5105;

d. carry out a minimum of one medical lock transfer with main RCC;

e. carry out a minimum of one personnel lock transfer with main RCC; and

f. ensure all applicable logs and records are complete.

4. Time 30 x 50 minute periods:

5. Method/ Media:

a. demonstration - 6 periods
b. practical - 13periods; and
c. review/ EC - 11 periods

6. References: A5 and Main RCC SOPs.

7. Training Aids: Main RCC and support systems

8. Test Details:

a. Pass mark is SAT on EC 401.3 checklist; and
b. 85 % on written exam.

9. Remarks: None

10. Teaching Points:

a. State RCC SOPs including;

(1) normal operation;
(2) emergency hyperbaric preparations;
(3) equipment emergencies;
(4) optimum manning;
(5) safe operating practise;
(6) rules for chamber ventilation including

(a) breathing air or HeO_2;
(b) breathing O_2 by mask;
(c) interrupted ventilation;
(d) O_2 monitoring equipment available; and
(e) O_2 overboard dump system installed.

(7) determination of air ventilation rates;
(8) chamber medical kits;
(9) chamber preparation;
(10) chamber operating procedures;
(11) tending the patient;
(12) locking in and out; and
(13) chamber dive records.

b. Explain emergency hyperbaric treatment, general preparation IAW ref A5 including;

(1) manning;
(2) equipment; and
(3) access.

c. conduct a demonstration dive with the main RCC and the SUBSAR RCC, to include:

(1) pre/post-dive checks;
(2) diving SOPs;
(3) medical lock transfer;
(4) personnel lock transfer; and
(5) checklists/ logs/ records;

c. emphasize importance of maintaining:

(1) ascent/descent rates;
(2) communications;
(3) stop depths;
(4) ventilation rates; and
(5) O_2 safety precautions;

EC 401.3

PART I – ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Main RCC

2. Support personnel.

ARRANGEMENTS AND PROCEDURES

3. The officer will operate the RCC, under supervision IAW classroom lecture, instructor demonstration, ref A5 and RCC SOPs.

4. References are allowed; assistance is denied.

5. Pass-mark is a grading of SAT on EC 401.3 checklist.

6. Time required: as per instructor dive profile

PART II - INSTRUCTIONS TO BE GIVEN TO THE TRAINEE

1. This EC will demonstrate your ability to operate the main RCC under supervision.

2. References are allowed; assistance is denied.

3. Pass-mark is a grading of SAT on EC 401.3 checklist

4. Time allowed: as per instructor profile.

PART III - ENABLING CHECKNOTES. Nil.

PART IV – CHECKLIST.

NAME: DATE;

TASK	S	U	REMARKS
Conduct pre/post dive routines IAW SOPs			
Establish comms with inside tender			
* Diving steps a. patient on O_2, standby to dive, thumbs up b. diving c. start clocks; and d. dives 25 fpm (+/- 2 fpm)			
Stop (ear problems) a. standby to come up 5 ft b. come up 5 ft c. try to clear, OK d. standby to dive, thumbs up e. diving; and f. @ 60 ft inform "on bottom", start O_2 clock			
* Maintain 60 ft (+/_ 1 ft)			
* Transfer article using medical lock			
(Air break) off O_2 . stop O_2 clock			
On O_2 ,start O_2 clock			
(O_2 hit) off O_2 stop O_2 clock			
Back on O_2 start O_2 clock			
Standby to travel to 30 ft @ 1 fpm			
* Travel 1 ft/min (+/- 10 sec for 5 mins)			
* @ 30 ft, maintain depth (+/- 1 ft)			
Lock in spare tender, log time			
* (O_2 alarm) vent chamber and maintain depth (+/- 1 ft) for 2 mins			
Standby to travel to surface thumbs up			
* travel to surface @ 60 fpm (+/- 2 fpm). stop clocks all logs complete			

1. Performance Operate DUOCOM transportable RCC

2. Conditions;

   a. Given;

      (1) DUOCOM RCC and support system;
      (2) SUBSAR/ Main RCC and support systems
      (3) support team;
      (4) references;
      (5) logs and records
      (6) applicable checklists;
      (7) stop watches;
      (8) dive profile; and
      (9) supervision.

   b. Denied; supervision

   c. Environmental N/A

3. Standard The officer shall;

   a. state DUOCOM RCC SOPs IAW ref A5

   b. operate the DUOCOM RCC including pre-post dive routines IAW ref A23, SOPs and checklists;

   c. operate DUOCOM in the following modes;

      (1) open circuit; and
      (2) semi-closed circuit.

   d. maintain;

      (1) required descent and ascent rates;
      (2) communications;
      (3) stop depths, to within +/- 1 fsw; and
      (4) required ventilation rate

   e. as a member on a team, carry out TUP procedures without RCCs being pressurized; and

   f. ensure all applicable logs and records are complete.

4. Time. 28 x 50 min periods

5. Method/media

   a. lecture - 5 periods
   b. demonstration - 4 periods
   c. practical - 10 periods
   d. review/ EC - 9 periods

6. References. A5 A23 A30 and Main/SUBSAR RCC SOPs for TUPs

7. Training Aids.

a. DUOCOM
b. Main/SUBSAR RCC
c. DUOCOM manual (A23)
d. Main/SUBSAR RCC SOPs

8. Test Details.

a. pass mark is SAT on checklist; and
b. 80 % on written exam.

9. Remarks.

a. two instructors should be available to allow trainees maximum practical time utilizing two DUOCOMs concurrently

b. Demonstration TUP (para 3d) to be carried out by qualified pers

c. tour of YDT 12 RCC system to be conducted at appropriate time.

d. recommend a tour of Victoria General Hospital RCC to be conducted if time permits.

10. Teaching Points.

a. state DUOCOM RCC SOPs including

(1) characteristics;
(2) system description;
(3) SOPs; and
(4) safety.

b. demonstrate DUOCOM pre-transportation checks IAW ref A23, part 4, sec 2 para 1-6 and checklist and fig 4-2-1 (6 sheets);

c. demonstrate DUOCOM support equipment muster IAW checklist(fig 4-2-2 (2 sheets);

d. demonstrate DUOCOM dive procedures IAW ref A23, part 4, sec 2 para 7-13 and checklist and fig 4-2-3 (8 sheets)including;

(1) pressurization system (step 1-6);
(2) BIBS (step 7-10);
(3) internal inspection (step 11-16);
(4) patient handling (step 17-21)
(5) chamber pressurization;

(a) semi-closed op (step 22a)
(b) open circuit op (step 22b)

(6) chamber depressurization;

(a) semi-closed op (step 23a)
(b) open circuit op (step 23b)

(7) medical lock op (step 24a)

e. demonstrate TUP procedures IAW ref A23;

f. demonstrate DUOCOM shut down and secure procedures IAWref A23; and

g. emphasize importance of maintaining;

(1) ascent/ descent rates;
(2) communications;
(3) stop depths;
(4) ventilation rates; and
(5) O_2 safety precautions.

EC 401.4

PART 1 - ADMINISTRATIVE DETAILS.

MATERIAL AND FACILITIES REQUIRED

1. DUOCOM RCC

2. Support personnel;

ARRANGEMENTS AND PROCEDURES.

3. the trainee will operate the DUOCOM RCC under supervision IAW classroom lecture, instructor demonstration, ref A5 and RCC SOPs.

4. The trainee is to be supplied with applicable references, Sops, records, support team and dive profiles.

5. Pass mark is SAT on checklist.

6. Time - as required by instructor dive profile.

7. Assistance is denied.

PART II - INSTRUCTIONS GIVEN TO TRAINEE.

1. This EC will demonstrate your ability to operate the DUOCOM RCC under supervision.

2. All necessary references, SOPs, records, equipment and support pers will be available.

3. Pass mark is SAT on checklist.

4. Time - as required by instructor dive profile.

5. Assistance is denied.

PART III - ENABLING CHECKNOTES. None.

PART IV CHECKLIST EC 401.4

NAME DATE

TASK	S	U	REMARKS
Conducts pre/post dive routines IAW SOPs			s=sat,u=unsat
Ensures patient and tender wear chamber clothing or clean clothing only			
* Dives the RCC in the mode specified by instructor.			
* Maintains descent rate specified by instructor (+/- 2 fpm)			
* Switches to an alternate mode			
Transfers article through medical lock.			
Ascends to 30 ft at rate specified by instructor			
* Maintain depth (+/- 1 fsw)			
* Switches from primary air/O2 to mating or external air/O2			
Surfaces at rate specified by instructor (+/- 2 fpm)			

a. * - denotes critical check

b. Trainee must score a SAT on all critical checks to achieve an overall grade of SAT

c. A scoring of UNSAT on two non-critical checks will be reviewed by the Senior instructor/ trg CPO to determine the effect on the overall grade.

EO 401.5

1. Performance: Supervise RCC operations

3. Conditions:

a. Given:

1. dive team;
2. dive profile;
3. main/SUBSAR RCC;
4. air and 0_2 supply;
5. support personnel/equipment;
6. equipment/system failure/breakdown scenarios;
7. references; and
8. supervision for safety.

b. Denied. assistance

c. Environmental. N/A

3. Standard: The trainee shall, IAW references and SOPs:

a. supervise an RCC dive including;

(1). a Main RCC dive; and
(2) a DUOCOM dive including a TUP

b. during the dive be tested orally and practically on a minimum of two emergency responses IAW ref A5.

4. Time: 24 x 50 minute periods:

5. Method/ Media:

a. practical/ performance - 14 periods; and
b. review/ EC - 10 periods

6. Reference: A5. A23, RCC SOPs

7. Training Aids:

a. DUOCOM RCC;
b. Main RCC;
c. manuals (A5 A31); and
d. RCC SOPs.

8. Test Details: Pass mark is SAT on checklist

9. Remarks: This EO should be performed concurrently with EO 401.3 and .4 to maximize effective training time.

10. Teaching Points: There are no additional knowledge required to support this EO. Carry out IAW EO 401.3

EC 401.5 SUPERVISE RCC OPERATIONS

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERAL AND FACILITIES REQUIRED.

1. RCC

2. Support pers

ARRANGEMENTS AND PROCEDURES

2. The trainee will supervise an RCC dive including a TUP and a minimum of two emergency scenarios. the instructor will provide supervision for safety..

3. References are allowed, assistance is denied.

4. Pass-mark is a grading of SAT on checklist.

5. Time required: as per instructor profiles

PART II - INSTRUCTIONS TO BE GIVEN TO THE TRAINEE

1. This EC will demonstrate your ability to supervise an RCC dive including a TUP and emergency responses.

2. References are allowed; assistance is denied.

3. Pass-mark is a grading of SAT on checklist.

4. Time allowed: as per instructor profiles.

PART III ENABLING CHECKNOTES. Nil

PART IV - EC 401.5 CHECKLIST

DATE_____

NAME_____

TASK	S	U	REMARKS
1. Pre-dive checks:	a		S=Sat, U=Unsat
a. *Pressure to console, FS S.	b		
b. *Air, HeO_2, O_2 to BIBS	c		
c. Check comms.	d		
d. Sufficient BIBS and headset (plus spare)	e		
e. *Clocks, charts, pencils, chamber suits.			
2. Pre-dive briefing:	a		
a. *Type of treatment (table, depth BT).	b		
b. Air, HeO_2, O_2 pressures	c		
c. *Console (rate of travel, vent).	d		
d. Charts (timekeeper 1 min warning for stops and travelling).	e		
e. Tenders (famil with BIBS).	f		
f. Ensure tenders clean.	g		
g. *Emergency procedure (lost gas, comms O_2symptoms).			
3. Chamber dive:	a		
a. *Ensure safe conduct of dive.	b		
b. Maintain control of dive team.	c		
c. *Correctly responds to emergency scenarios	d		
d. Maintain post for best control of operations			

NOTE: a. *Denotes critical check.

b. Trainee must score a SAT on all critical checks to achieve an overall grading of SAT for EC 404.1.

c. A scoring of UNSAT on four out of eight non-critical will reviewed by training staff to determine the effect on the overall grade

PART I - ADMINISTRATIVE INSTRUCTIONS.

MATERIAL AND FACILITIES REQUIRED.

1. RCCs

2. treatment tables

3. references.

ARRANGEMENTS AND PROCEDURES

4. This PC is considered successfully completed when all supporting EOs have been successfully completed.

PART II - INSTRUCTIONS GIVEN TO TRAINEE.

1. This PC is considered successfully completed when all supporting EOs have been successfully completed.

PART II - PERFORMANCE CHECKNOTES. None.

PART IV - PERFORMANCE CHECKLIST

EC	S	U	REMARKS
Operate main RCC			
Operate DUOCOM			
perform as a member of TUP team			
Supervise RCC ops			

1. Performance: Apply diving physics

2. Conditions:

   a. Given. nil

   b. Denied;

      (1) assistance; and
      (2) references.

   c. Environmental. N/A

3. Standard: The trainee shall apply diving physics IAW ref A2 chap 2 by:

   a. defining the fundamental laws of diving physics, and
   b. calculating diving physics formulae.

4. Timing: 12 x 50 minute periods:

5. Method/Media:

   a. Lecture - 7 periods;
   b. home assignment - 2 periods
   c. EC review - 3 periods

6. References: A2

7. Training Aids:

   a. OHPs,
   b. Multi media systems
   c. whiteboard
   d. homework assignments; and
   e. calculators

8. Test Details: Pass mark is 80 %

9 Remarks: This EO is conducted at DCIEM, ASU Toronto.

10. Teaching Points:

   a. Explain diving physics IAW ref A1, chap 2, to include:

      (1) basic concepts (sect 1, art 2101),
      (2) matter and energy (sect 2), including:

         (a) matter (art 2201),
         (b) three states of matter (art 2202),
         (c) water (art 2203),
         (d) common gases (art 2204),
         (e) six forms of energy (art 2206), and
         (f) energy in diving (art 2207),

(3) pressure (sect 3), including:

(a) pressure in general (art 2301),
(b) types of pressure (art 2302),
(c) liquid pressure (art 2303),
(d) kinetic theory of gases (art 2304), and
(e) absolute temperature (art 2305),

(4) gas laws (sect 4), including:

(a) gas laws in general (art 2401),
(b) Boyle's Law (art 2402),
(c) Charles Law (art 2403), and
(d) General Gas Law (art 2404),

(5) gas mixtures and gas absorption in liquids (sect 5), includ-ing:

(a) gas mixtures (art 2501),
(b) Dalton's Law of Partial Pressures (art 2502),
(c) surface equivalent (art 2503),
(d) gas diffusion (art 2504),
(e) gas absorption in liquids (art 2505),
(f) Henry's Law (art 2506), and
(g) Henry's Law and decompression (art 2507),

(6) air humidity (sect 6), including:

(a) terms and definitions (art 2601),
(b) effects of compression and expansion (art 2602), and
(c) regulator freeze-up (art 2603),

(7) buoyancy (sect 7), including:

(a) Archimedes’s Principle (art 2701),
(b) buoyancy control (art 2702),
(c) individual buoyancy variations (art 2703), and
(d) laws of flotation (art 2705),

(8) light, heat and sound underwater (sect 8), including:

(a) light (art 2802),
(b) refraction of light (art 2803),
(c) underwater visibility (art 2804),
(d) sound (art 2805),
(e) helium speech (art 2806),
(f) heat (art 2807),
(g) heat transfer (art 2808),
(h) heat transfer from the body (art 2809),
(j) thermal protection (art 2810),
(k) heat loss in a gas environment (art 2811), and
(l) underwater mobility (art 2812), and

(9) units of measurement (annex (E)-2A), including:

(a) units of measurement in general (para 1),
(b) metric system (para 2-6),
(c) Fahrenheit and Celsius temperature scales (para 7-9),
(d) absolute temperature scales (para 10),

(e) barometric pressure (para 11-14), and
(f) use of conversion factors (para 15-19), and

b. demonstrate, by use of examples, calculations of diving physics formulae, to include:

(1) water pressure (art 2303, para 3),
(2) absolute temperature (art 2305, para 2),
(3) Boyle's Law (art 2402, para 8),
(4) Charles's Law (art 2403, para 2),
(5) General Gas Law (art 2404, para 1),
(6) Dalton's Law of Partial Pressure (art 2502, para 4),
(7) surface equivalent (art 2503, para 2),
(8) Archimedes’s Principle (art 2701, para 1-4),
(9) metric system (annex (E)-2A, para 2-6), and
(10) Fahrenheit and Celsius temperature conversions (annex (E)-2A, para 7-9).

EO 402.2

1. Performance: Explain diving physiology and medicine

2. Conditions:

   a. Given. nil

   b. Denied.

      1. assistance
      2. references

   c. Environmental. N/A

3. Standard: The trainee shall explain diving physiology and medicine IAW ref A2, chap 3, to include:

   a. physiology;
   b. direct effects of pressure;
   c. indirect effects of pressure;
   d. respiratory problems;
   e. temperature regulation;
   f. miscellaneous hazards; and
   g. medical problems and diving.

4. Time: 27 x 50 minute periods:

5. Method/ Media:

   a. Lecture - 23 x 50 min periods
   b. EC - 4 x 50 min periods

6. References: A2

7. Training Aids:

   a. multi-media systems;
   b. printed aids;
   c. laptop computer;
   d. whiteboard;
   e. videos
   f. anatomy charts; and
   g. DCIEM handouts.

8. Test Details: Pass mark is 80 %

9. Remarks:

   a. Training conducted at DCIEM ASU Toronto; and
   b. Practical application of knowledge gained from this EO will be carried out in POs

10. Teaching Points

a. Explain diving physiology and medicine including all applicable ailment signs, symptoms and treatments IAW ref A2, chap 3, to include:

(1) the human body (sect 1, art 3101);
(2) physiology (sect 2), including:

(a) circulatory system (art 3202);
(b) heart (art 3203);
(c) blood pressure (art 3204);
(d) blood (art 3205);
(e) respiratory (art 3206);
(f) respiratory terminology (art 3207);
(g) pulmonary anatomy (art 3208);
(h) respiratory activity (art 3208);
(j) control of respiration (art 3210);
(k) mechanics of respiration (art 3211);
(l) resistance to breathing (art 3212);
(m) ventilation and perfusion (art 3213);
(n) gas exchange (art 3214);
(o) oxygen consumption (art 3215);
(p) CO_2 production and elimination (art 3216);
(q) 12 pairs of cranial nerves
(r) autonomic nervous system (art 3218);
(s) hormones (art 3219);
(t) muscular-skeletal system (art 3220);
(u) metabolism (art 3221);

(3) direct effects of pressure (sect 3),including:

(a) effects of very high pressure on living cells (art 3302);
(b) effects of unequal pressure (art 3303);
(c) ears (art 3304);
(d) inner ear barotraumas (art 3305);
(e) squeeze (chap 2, sect 4, art 2405);
(f) sinuses (chap 3, sect 3, art 3306);
(g) teeth (art 3307);
(h) gut (art 3308);
(j) face and skin (art 3309);
(k) lungs (art 3310);
(l) air embolism (art 3311);
(m) treatment of air embolism (art 3312);
(n) Pneumothorax and mediastinal emphysema (art 3313); and
(o) bleeding with barotraumas (art 3314);

(4) indirect effects of pressure (sect 4), including:

(a) oxygen toxicity (art 3402);
(b) lung oxygen toxicity (art 3403);
(c) nervous system oxygen toxicity (art 3404);
(d) symptoms of nervous system oxygen toxicity (art 3405);
(e) inert gas narcosis (art 3406);
(f) basis for decompression schedules (art 3407);
(g) decompression sickness (art 3408); and
(h) dysbaric osteonecrosis (art 3409);

(5) respiratory problems (sect 5), including:

(a) hypoxia (art 3502);
(b) prevention of hypoxia (art 3503);
(c) carbon dioxide excess (art 3504);
(d) asphyxia, suffocation and strangulation(art 3505, para 1-2);
(e) control of massive bleeding (art 3506);
(f) breath-holding (art 3507);
(g) hyperventilation (art 3508);
(h) carbon monoxide poisoning (art 3509); and
(j) treatment of carbon monoxide poisoning (art 3510);

(6) temperature regulation (sect 6), including:

(a) effects of temperature (art 3601);
(b) temperature regulation (art 3602);
(c) effects of heat loss (art 3603);
(d) hypothermia (art 3604);
(e) surface exposure to cold (art 3605); and
(f) overheating in water (art 3606);

(7) miscellaneous hazards (sect 7), including:

(a) drowning (art 3701);
(b) pathophysiology of drowning (art 3702);
(c) treatment of near drowning (art 3703);
(d) overexertion (art 3704);
(e) seasickness (art 3705);
(f) fainting (art 3706);
(g) physical injury (art 3707);
(h) underwater blast injuries (art 3709);
(j) burns (art 3710);
(k) electric shock (art 3711);
(l) hazardous marine life (art 3712); and
(m) nuclear radiation (art 3713); and

(8) medical problems and diving (sect 8), including:

(a) fitness to dive (art 3802);
(b) upper respiratory infections (art 3803);
(c) external eye and ear infections (art 3804);
(d) fungus infections (art 3805);
(e) skin conditions (art 3806);
(f) loss of consciousness (art 3807);
(g) obesity (art 3808);
(h) diet and alcohol (art 3809); and
(j) drugs and diving (art 3810 and annex (E)-3B).

EC 402.2

PART I – ADMINISTRATIVE INSTRUCTIONS

1. Classroom.

ARRANGEMENTS AND PROCEDURES

2. The trainee will complete a written test on diving physiology and medicine.

3. Assistance and references are denied.

4. Pass mark is 80%.

5. Time required: 3 period.

PART II - INSTRUCTIONS TO BE GIVEN TO THE TRAINEE

1. This EC will confirm your knowledge of diving physiology and medicine.

2. Assistance and references are denied.

4. Pass-mark is 80%.

5. Time allowed: 3 period.

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE
PHYSIOLOGY	circulatory system	5
	respiratory system	5
	nervous system	5
	muscular-skeletal system	5
	SUB-TOTAL	**20**
DIRECT EFFECT OF PRESSURE	ears, inner ear barotraumas	3
	squeeze	3
	sinuses	3
	teeth, gut, face and skin, lungs	2
	Pneumothorax and mediastinal emphysema	4
	SUB-TOTAL	**25**
INDIRECT EFFECTS OF PRESSURE	oxygen toxicity	4
	inert gas narcosis	2
	decompression sickness	14
	SUB-TOTAL	**20**
RESPIROTORY PROBLEMS	hypoxia	2
	CO_2 excess	3
	breath holding	2
	hyperventilation	2
	CO poisoning	3
	SUB-TOTAL	**12**
TEMERATURE REGULATION	regulation	2
	effects of heat loss	2
	hyperthermia	4
	SUB-TOTAL	**9**
MISCELLANEOUS HAZARDS	drowning	2
	overexertion	1
	u/w blast injuries	2
	electric shock	2
	SUB-TOTAL	**7**
MEDICAL PROBLEMS AND DIVING	fitness to dive	1
	infections	2
	loss of consciousness	1
	obesity	1
	diet, alcohol and drugs	2
	SUB-TOTAL	**7**

PART IV – ENABLING CHECKLIST N/A

EO 402.3

1. Performance: Explain hyperbaric treatment tables

2. Conditions:

   a. Given. Nil

   b. Denied

      (1) references
      (2) assistance

   c. Environmental. N/A

3. Standard: The trainee shall explain hyperbaric tables/ oxygen treatment tables, including:

   a. 5;
   b. 6 and options;
   c. 6A and options;
   d. 7;
   e. 8;and
   f. other alternate treatment tables.
   g. explain

4. Time: 9 x 50 min periods:

5. Method/Approach:

   a. Lecture - 7 periods; and
   b. EC - 2 periods

6. References: A5.

7. Training Aids:

   a. multi-media systems;
   b. printed aids;
   c. laptop computer; and
   d. whiteboard

8. Test Details: Pass mark is 80 %

9. Remarks: This EO is taught by DRDC, ASU Toronto.

10. Teaching Points:

a. Explain oxygen treatment tables, to include:

(1) Table 5 and use;
(2) Table 6 and use;
(3) Table 6 modified and use;
(4) Table 6A and use;
(5) Table 6A modified and use;
(6) Table 7;
(7) Table 8;
(8) other alternate Treatment Tables;

b. Explain why oxygen is more beneficial than air treatments

EC 402.3 EXPLAIN HYPERBARIC TREATMENT TABLES

PART I – ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom

ARRANGEMENTS AND PROCEDURES

2. The trainee will complete a written test on hyperbaric treatments tables IAW classroom lecture.

3. Assistance and references are denied.

4. Pass-mark is 80%.

5. Time required: 1 period

PART II - INSTRUCTIONS TO BE GIVEN TO THE TRAINEE

1. This EC will confirm your knowledge of hyperbaric tables.

2. Assistance and references are denied.

3. Pass-mark is 80%.

4. Time allowed: 1 period

PART III – ENABLING CHECKNOTES.

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE (%)
OXYGEN TREATMENT	Table 5	10%
	Table 6	20%
	Table 6 modified	20%
	Table 6A	20%
	Table 6A modified	20%
	Table 7	5%
	Alternate treatment tables	5%

EO 402.4

1. Performance; Explain diving ailments requiring hyperbaric treatment.

2. Conditions.

   a. Given. classroom.

   b. Denied. assistance and references.

   c. Environmental. N/A

3. Standard. The trainee shall explain diving ailments requiring hyperbaric treatment IAW ref A5, Chap 3,sec 2 and Ref A3, Chap 2,sec 6 to include signs and symptoms for:

   a. decompression illness; and
   b. omitted decompression.

4. Time. 11 x 50 mins periods

5. Method/ Media.

   a. lecture - 9 periods; and
   b. EC - 2 periods

6. References - A2 to A5, A26

7. Training Aids.

   a. multi-media systems;
   b. laptop computer; and
   c. printed aids

8. Test Details. Pass mark is 85 %

9. Remarks. This EO is conducted at DCIEM, ASU Toronto

10. Teaching Points.

    a. explain and demonstrate rapid neurological exam;
    b. explain and demonstrate IV's;
    c. explain and demonstrate obtaining vital signs; and
    d. explain diving ailments requiring recompression, including;

       (1) DCS;
       (2) pulmonary over-inflation syndrome; and
       (3) omitted decompression.

EC 402.4

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED.

1. Classroom.

ARRANGEMENTS AND PROCEDURES.

2. The trainee will complete a written exam on diving ailments requiring recompression IAW classroom lecture.

3. Assistance and references.

4. Pass mark is 85 %

5. Time required - 1 period

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of diving ailments requiring recompression.

2. Assistance and references are denied.

3. Pass mark is 85 %.

4. Time required 1 period.

PART III - ENABLING CHECKNOTES.

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE
DECOMPRESSION SICKNESS	DCS	30
	omitted decompression	6
	vital signs	
PULMONARY OVER-INFLATION SYNDROME	gas embolism	28
	mediastinal emphysema	12
	subcutaneous emphysema	12
	Pneumothorax	12

ORIGINAL

NOTE: Standards statement does not reflect everything that is in "Teaching Points" Recommend using QL 6B QSP.

EO 402.4 **(CLDO)**

1. Performance; Explain diving ailments requiring hyperbaric treatment.

2. Conditions.

   a. Given. classroom.

   b. Denied. assistance and references.

   c. Environmental. N/A

3. Standard. The trainee shall explain diving ailments requiring hyperbaric treatment IAW ref A5, Chap 3,sec 2 and Ref A3, Chap 2,sec 6 to include signs and symptoms for:

   a. decompression illness; and
   b. omitted decompression.

4. Time. 11 x 50 mins periods

5. Method/ Media.

   a. lecture - 9 periods; and
   b. EC - 2 periods

6. References - A2 to A5, A26

7. Training Aids.

   a. multi-media systems;
   b. laptop computer; and
   c. printed aids

8. Test Details. Pass mark is 85 %

9. Remarks. This EO is conducted at DCIEM, ASU Toronto

10. Teaching Points.

    a. explain and demonstrate rapid neurological exam;
    b. explain and demonstrate IV's;
    c. explain and demonstrate obtaining vital signs; and
    d. explain diving ailments requiring recompression, including;

       (1) DCS;
       (2) pulmonary over-inflation syndrome; and
       (3) omitted decompression.

EC 402.4

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED.

1. Classroom.

ARRANGEMENTS AND PROCEDURES.

2. The trainee will complete a written exam on diving ailments requiring recompression IAW classroom lecture.

3. Assistance and references.

4. Pass mark is 85 %

5. Time required - 1 period

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of diving ailments requiring recompression.

2. Assistance and references are denied.

3. Pass mark is 85 %.

4. Time required 1 period.

PART III - ENABLING CHECKNOTES.

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE
DECOMPRESSION SICKNESS	DCS	30
	omitted decompression	6
	vital signs	
PULMONARY OVER-INFLATION SYNDROME	gas embolism	28
	mediastinal emphysema	12
	subcutaneous emphysema	12
	pneumothorax	12

EO 402.5

1. Performance. Supervise emergency treatment of diving ailments and underwater blast injuries.

2. Conditions.

   a. Given

      (1) dive team;
      (2) first aid equipment;
      (3) RCC;
      (4) diving scenario with an emergency situation;
      (5) references/ SOPs; and
      (6) supervision.

   b. Denied. assistance.

   c. Environmental. N/A

3. Standard. The trainee shall supervise the treatment for a diving ailment requiring and/or underwater injury including:

   a. identifying the ailment and/or injury;
   b. diagnosis both the diving ailment and/or the injury, using the rapid neurological exam;
   c. selecting definitive treatment IAW ref A5, chap 3 and ref A3, chap 2;
   d. initiate and carryout appropriate treatment scenario IAW refs.
   e. emergency scenarios, to include;

      (1) one - non hyperbaric treatment; and
      (20 two - hyperbaric treatments

4. Time. 35 x 50 min periods

5. Method/Media.

   a. lecture/ demo/ performance - 19 periods
   b. EC - 16 periods

6. References. A3, A5.

7. Training Aids.

   a. multi-media systems;
   b. printed aids; and
   c. laptop computer.

8. Test Details. Pass mark is SAT on checklist.

9. Remarks.

   a. practical assessment of hyperbaric diving emergencies will be conducted at DCIEM, ASU Toronto; and
   b. practical assessment of non-hyperbaric diving emergencies should be conducted in conjunction with POs.

10. Review immediate emergency care, including;

a. first aid;
b. position;
c. fluids;
d. oxygen;
e. transportation;
f. recompression in water; and
g. recompression in RCC.

EC 402.5

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILIITES REQUIRED

1. RCC

2. Support personnel

ARRANGEMENTS AND PROCEDURES

3. The trainee will supervise emergency treatment of diving ailments and underwater injuries with the instructor providing supervision for safety.

4. Assistance is denied.

5. Pass mark is SAT on checklist

6. Time required is 35 x 50 min periods

PART II - INSTRUCTIONS GIVEN TO TRAINEE.

1. This EC will confirm your ability to supervise emergency treatment of diving ailments and underwater injuries.

2. Assistance is denied.

3. Pass mark is SAT on checklist

4. Time required is 35 x 50 min periods.

PART III - ENABLING CHECKNOTES. n/a

PART IV - ENABLING CHECKLIST. EC 402.5

NAME

DATE

TASK	S	U	REMARKS
* Properly identified the u/w injury or ailment			S = Sat, U = Unsat
Performs rapid neurological exam			
* Proper diagnosis of u/w injury or ailment			
* Selected proper treatment for ailment of u/w injuries			
* Initiated and carried out appropriate treatment			
Completed one non-hyperbaric treatment			
Completed two hyperbaric treatments			
* Observes all safety precautions			

a. * denotes critical check.
b. Trainee must score a SAT on all critical checks to achieve and overall grading of SAT.
c. A scoring of UNSAT on two non-critical checks will be reviewed by the senior instructor/trg CPO to determine the effect on the overall grade.

PC 402

PART I - ADMINISTRATIVE INSTRUCTIONS.

MATERIAL AND FACILITIES REQUIRED.

1. RCC

2. Support personnel.

ARRANGEMENTS AND PROCEDURES.

3. The trainee shall supervise emergency treatment of diving ailments.

4. Pass mark is SAT on the checklist.

PART II - INSTRUCTIONS GIVEN TO TRAINEE.

1. This PO will confirm your ability to supervise emergency treatments of diving ailments.

2. Pass mark is SAT on checklist

PART III - PERFORMANCE CHECKNOTES.

1. These checks will be conducted at DCIEM with the exception of those that will be conducted IAW applicable POs.

PART IV - PERFORMANCE CHECKLIST. PC 402

NAME

DATE

EC	S	U	REMARKS
Apply diving physics			
Explain diving physiology and medicine			
* Explain hyperbaric treatment tables			
* Explain diving ailments requiring hyperbaric treatment			
* Supervise emergency treatment of diving ailments and u/w injuries.			

PO 403 Conduct basic EOD procedures.

EO 403.1

Related EOs taught through CFSEME CFB Borden.

EO 404.1

1. Performance. Explain CABA operations

2. Conditions:

   a. Given; classroom

   b. Denied; Assistance and references.

   c. Environmental. N/A

3. Standard. The trainee shall explain CABA operations IAW refs including;

   a. rules and regulations; and
   B. shot-line diving procedures.

4. Time. 2 x 50 min periods

5. Method/Media

   a. lecture - 1 period
   b. EC - 1 period

6. References. A2,A3

7. Training Aids. None

8. Test Details. Pass mark is 80 %

9. Remarks. none

10. Teaching Points.

    a. Diving responsibilities of the following NDHQ and command offices;

       (1) CMDO/DGMDO/DMOPR;
       (2) CADO/DGADO/DAOT;
       (3) DCDS/DGMEO/DMEO;
       (4) CEM/DGMEM/DMEE;
       (5) SURGEN/DPM; and
       (6) SODMCM.

    b. Clearance diving officers and clearance divers operating and supervisory limits for CABA.

    c Standby diver, including;

       (1) states of readiness; and
       (2) circumstances that require the highest states of readiness.

    d. Minimum and maximum number of divers to be employed at one time.

    e. Operational depth limits for CABA.

f. Requirements for a RCC for all types of diving, dependant upon depth.

g. Who is empowered to authorize exceeding specified depth limits.

h. The difference between marked swimming and float diving.

i. Precedence of CFAO 43-2 as a primary administrative order dealing with the CF diving organization, responsibilities and administrative procedures.

j. Diving incident/ accident reports including;

(1) 24 hour time limit for reporting;
(2) the necessity to report;
(3) investigation of life support equipment after a diving incident or accident;
(4) types of incidents/ accidents for which a report is mandatory; and
(5) the specific action to be taken on recovery.

k. Flying after diving.

l. Shot line diving equipment.

m. Shot line diving SOPs including;

(1) lazy shot procedures; and
(2) decompression procedures.

n. Special signals.

o. Explain Table 1 to complete decompression schedule.

EC 404.1

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED.

1. Classroom.

ARRANGEMENTS AND PROCEDURES.

2. The trainee will complete a written exam on CABA diving operation IAW classroom lecture.

3. Assistance and references are denied

4. Pass mark is 80 %

5. Time required - 1 period.

PART II - INSTRUCTIONS GIVEN TO TRAINEE.

1. This EC will confirm your knowledge of CABA and diving rules and regulations

2. Assistance and references are denied.

3. Pass mark is 80 %

4. Time allowed - 1 period.

PART III - ENABLING CHECK NOTES

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE
DIVING OPERATIONS	CABA operations and supervision	35
	Standby diver	7
	Number of diver employed	3
	Requirements for an RCC	7
	Depth/time limitations	10
	Flying after diving	6
ADMINISTRATIVE	Command responsibilities	5
	CFAO 43-2	12
	Diving incident/accident reports	15

PART IV - ENABLING CHECKLIST. N/A

EO 404.2

1. Performance: Conduct CABA operations

1. Conditions:

a. Given:

(1). CABA;
(2). dive team and associated equipment;
(3). supervision; and
(4). references.

b. Denied, assistance

c. Environmental. in existing weather conditions, day or night

3. Standard: The trainee shall:

a. Carry out CABA dives to include:

(1) a minimum of 3 no-decompression dives deeper than 50 fsw;
(2) a minimum of one in-water decompression dive(using lazy shot IAW ref
(3) the lazy shot method shall be carried out on a minimum of one of these dives; and
(4) a minimum of one CABA dive shall be to within 20 fsw of the maximum operating depth.

b. carry out pre-post dive routines on the equipment; and
c. apply all safety precautions.

4. Time 19 x 50 minute periods:

5. Method/ Media:

a. Lecture - 2 periods;
b. Performance - 13 periods; and
c. EC - 4 periods.

6. References: A3, A51.

7. Training Aids: Equipment IAW para 2 (conditions)

8 Test Details: Pass mark is SAT on checklist

9. Remarks:

a. the trainee has received some prior CABA training on a STD course;

b. this EO is conducted in conjunction with EO 404.3 (supervise CABA ops);

c. EO 407.5 may be done concurrently with this EO.

10. Teaching Points.

a. the requirements for a worked up diver, supporting the dive team / equipment and RCC

b. the procedures for the use of the shot line, lazy shot and associated equipment; and

c. carryout decompression while drifting free with the use of inflatable boats.

EC 404.2

PART I - ADMINISTRATIVE INSTRUCTIONS.

MATERIAL AND FACILITIES REQUIRED.

1. Training area

2. Safety equipment

ARRANGEMENTS AND PROCEDURES

3. The trainee will perform dives using CABA IAW ref A3

4. References are allowed, assistance is denied.

5. Pass mark is SAT on checklist

6. Time required - 4 periods

PART II - INSTRUCTIONS GIVEN TO TRAINEE.

1. This EC will demonstrate your ability to perform dives using CABA and AGA in swim mode.

2. References are allowed, assistance is denied.

3. Pass mark is SAT on checklist.

4. Time required - 4 periods

5. Any safety violation will result in failure.

PART III - ENABLING CHECK NOTES. N/A

PART IV - ENABLING CHECKLIST. EC 404.2

NAME

DATE.

TASK	S	U	REMARKS
Performs CABA pre dive routines			S-Sat, U=Unsat
* Carries out a minimum of 3 no - D dives			
* Carries out a minimum of 1 decompression dive using lazy shot method.			
* Dives to a maximum of 2/3 max op depth			
Communicates u/w using SCUBA phone			
Performs post dive routines			
* Applies all safety precautions			

a. * denotes critical check.
b. trainee must score a SAT on all critical checks to achieve an overall grading of SAT
c. a scoring of UNSAT on 2 non-critical checks will be reviewed by the senior instructor / trg CPO determine the effect on the overall grade.

EO 404.3

1. Performance. Supervise CABA operations

2. Conditions:

   a. Given:

      (1) CABA;
      (2) dive team;
      (3) diving equipment
      (4) supervision;
      (5) references, and
      (6) scenario

   b. Denied; assistance

   c. Environmental: in existing weather conditions, day and night

3. Standard. The trainee shall, IAW references, plan, conduct and supervise CABA:

   a. to within 20 fsw of the maximum depth authorized;
   b. a minimum of 3 no-D dives greater than 50 fsw; and
   c. a minimum of 1 simulated decompression dive using lazy shot decompression procedures, including

      (1) formulate a dive plan;
      (2) calculate decompression schedules;
      (3) perform decompression procedures;
      (4) carry out pre/ post dive procedures and planned maintenance;
      (5) maintain control of the dive team throughout the dive;
      (6) respond to diving emergencies;
      (7) completeness and accuracy of reports and returns; and
      (8) completeness and accuracy of CF incident/ accident report.

4. Time: 10 X 50 Min periods

5. Method/ Media.

   a. demonstration - 1 period;
   b. performance - 6 periods; and
   c. EC - 3 periods

6. References. A3, C6, C7

7. Training Aids. nil

8. Test details. Pass mark is SAT on checklist.

9. Remarks. EO 407.6 may be done concurrently with this EO.

10. Teaching Points.

    a. all dives should be no-D
    b. explain/ demonstrate lazy shot drifting decompression dives procedures.

EC 404.3

PART I - ADMINISTRATIVE INSTRUCTIONS.

1. Training area

2. Recovery equipment.

3. Safety equipment.

ARRANGEMENTS AND PROCEDURES

4. The trainee will supervise dives using CABA with the instructor providing supervision for safety.

5. Assistance is denied

6. Pass mark is SAT

7. Time required - 3 periods

8. EO 407.6 may be done concurrently with the EO

PART II - INSTRUCTIONS GIVEN TO TRAINEE.

1. This EC will demonstrate your ability to supervise dives using CABA

2. Assistance is denied.

3. Pass mark is SAT

4. Time required - 3 periods.

PART III - ENABLING CHECK NOTES. All dives are to be no-D dives.

PART IV – PC 404 SUPERVISE CABA DIVE

NAME_____ DATE_____

TASK	S	U	REMARKS
1.Dive planning (for personnel, equipment, transportation and alternatives, depth, U/W conditions).			S = SAT U = UNSAT
2.Briefing (concise, good definition of task etc., any dives in last 18 hours, in case of emergency).			
3. Inform QHM, local harbour authorities, ship, as required.			
4. If applicable, clears ship IAW Ref A1, A24.			
5. Support equipment complete (boat running, floats, lines, flags/lights (dive) up/on).			
6. Standby dressed in boat and checked prior to other divers entry.			
7. Supervisor has diver recall signals.			
8. Check divers prior to entry.			
9. When divers are ready to enter water supervisor is to: a. order entry, b. observe entry, c. organize teams in water, and d. order divers down.			
10. Ensure "left surface time" is recorded			
11. Position safety boat in a position of maximum observation for harbour traffic and divers safety.			
12. As divers reach surface : a. check divers conditions; b. ensure "reached surface time" is recorded; and c. Ensure divers remain under his positive control and direction from arriving on surface to exiting water.			
13. Secure boat, standby, flag/light			
14. Inform QHM, local harbour authorities, ship, as required.			
15. Debrief.			

PO 404 Supervise Self-Contained Breathing Apparatus (SCBA) Operations.

EO 404.4

1. Performance. Explain CCDA Operations.

2. Conditions.

a. Given:

(1) references;
(2) classroom;
(3) diving / MCM equipment; and
(4) technical support.

b. Denied: assistance and references.

c. Environmental: n/a.

3. Standard. Explain CCDA operations IAW references to include:

a. rules and regulations;
b. function and operation;
c. MCM procedures;
d. beach survey; and
e. GPS / hand-held sonar, ATPs.

4. Time. 8 x 50-minute periods.

5. Method/Media.

a. lecture / demonstration / discussion - 6 periods; and b. EC - 2

period. 6. References. A33 Book 1, A30, A31, A43, A44, A47, C6, C7.

7. Training Aids.

a. CCDA; b. equipment drawings; and c. OHP.

8. Test Details. Written test. Pass mark - 80%.

9. Remarks. Not used.

10. Teachings Points.

a. Explain the rules and regulations for diving in CCDA to include:

(1) minimum personnel required;
(2) diving responsibilities;
(3) proficiency of divers;
(4) MCM diving, including ATPs 6B and 24B;
(5) operating in vicinity of live ordnance:

(a) Zodiac; and
(b) use of divers; and

(6) action in the event of a diving accident.

b. Explain the function and operation of CCDA to include:

(1) counter lung;
(2) counter lung relief valve (CRV);
(3) buoyancy control valve (BCV);
(4) carbon dioxide absorbent canister;
(5) hoses, mouthpiece and mask;
(6) cylinders pressure and capacities;
(7) counter lung volume;
(8) C02 canister capacity;
(9) apparatus weight in air and seawater;
(10) additional supporting equipment;
(11) gas flasks and manifold valve assembly;
(12) flask reserve valve;
(13) pressure regulator;
(14) flow selector valve;
(15) bypass valve;
(16) harness;
(17) counterweights;
(18) backpack;
(19) closed / semi-closed circuit rebreather design;
(20) clearance diving applications;
(21) shallow water diving and swimming applications;
(22) depth limitations and gas mixture;
(23) decompression;
(24) gas endurance;
(25) canister endurance;
(26) oxygen breathing limitations;
(27) underwater endurance of apparatus;
(28) planned maintenance and calibration routines to include:

(a) explain the conditions for calibration and maintenance, to include:

i. frequency and numbering of all tests;
ii. minimum supply pressure;
iii. maintenance logs and record sheets;
iv. use of 02-compatible, special-purpose grease for lubrication;
v. maintainer qualification; and
vi. work area;

(b) explain pressure regulator planned maintenance and calibration (CCDA Test #2);

(c) explain by-pass valve planned maintenance and calibration (CCDA Test #3);
(d) explain flow selector valve assembly (including micro-metering valves) planned maintenance and calibration (CCDA Test #4);
(e) explain counter lung relief valve and buoyancy control valve planned maintenance and calibration (CCDA Test #5);
(f) explain the leak test (CCDA Test #6);
(g) explain bailout assembly planned maintenance and calibration (CCDA Test #7);
(h) explain soft seal replacement (CCDA Test #8);

(i) explain spherical flasks testing and planned maintenance (CCDA Test # 9) to include:

i. testing conducted by SRU;
ii. interior and exterior inspection;
iii. hydrostatic proof test;
iv. recording in serial numbered log; and
v. special equipment required;

(j) explain CCDA calibration panel (CCDA Test #10) to include:

i. pressure gauge re-calibration;
ii. flow meter testing;
iii. working pressure test; and
iv. oxygen cleaning requirements;

(k) explain the storage and care of CCDA, Art 5110; and

(l) explain the requirements for non-magnetic hygiene to include:

i. magnetic cleanliness; and
ii. use of the non-magnetic tool kit (CCDA);

(29) pre / post dive procedures to include:

(a) pre and post dive routines;
(b) pre-dive assembly;
(c) charging the canister;
(d) installing the mask / hose assembly;
(e) analyzing gases;
(f) gas supply and gauge checks;
(g) charging **CCDA** cylinders IAW ref A47
(h) handling the apparatus;
(i) leak test;
(j) CCDA pre-dive tag; and
(k) set preparation between dives;

(30) apply oxygen and mixed gas breathing theory to include:

(a) explain the theory of oxygen breathing to include:

i. depth limitations; and
ii. oxygen toxicity risks;

(b) explain the theory of mixture breathing to include:

i. maximum safe depth for a given gas mixture;
ii. oxygen content in the breathing loop;
iii. formula for percentage of oxygen in the breathing loop;
iv. correct safe mass flow for any mixture;
v. equivalent air depth (EAD);

(i). plodding; and
(ii) swimming

vi. calculations of endurance for**:**

(i) CCDA diving set IAW art 1109:

oxygen breathing; and
mixture breathing;

(ii) carbon dioxide absorbent IAW ref A30; and

(c) demonstrate, by use of examples, all formulas calculations for CCDA.

c. MCM procedures, IAW reference to include:

(1) hand-held GPS;
(2) hand-held sonar to include:

(a) describe the AN / PQS-2A Sonar Set to include:

i. general information (Sect I);
ii. operation (Sect II); and
iii. scheduled maintenance (Sect IV, Para 4-1 to 4-2);

(b) explain directions for a flooded sonar IAW USN EOD Equipment Note No. 8-87, 16 Apr 87
and No. 19-87, 01 Oct 87;

(c) briefly describe the Dukane Underwater Acoustic Locator System IAW ref A42, Appendix B to include:

i. description;
ii. operation - general;
iii. controls and connectors;
iv. assembly and operation from a boat;
v. assembly and operation by a diver;
vi. battery replacement; and
vii. cleaning;

(d) describe underwater Acoustic Beacons IAW ref A42, Sect I and III to include:

i. physical characteristics;
ii. beacon signal;
iii. beacon cleaning; and
iv. battery replacement;

(e) briefly describe the FEREX 4.021 Search Instrument IAW ref A43 to include:

i. general information;
ii. description ;
iii. batteries;
iv. assembly and operating instruction; and
v. search techniques; and

(f) briefly describe the METEX 4.125 Mine and Metal Search Instrument IAW ref A58 to include:

i. introduction;
ii. construction;
iii. connecting-up;
iv. battery installation;

v. setting-up;
vi. search techniques;
vii. operator's Do's and Don'ts; and
viii. trouble-shooting; and

(3) mine field discipline.

d. Beach survey:

(1) explain beach survey terminology to include:

(a) type of breakers; (b) beach terminology; and (c) tide terminology.

(2) describe beach survey operations and procedures IAW references to include:

(a) tasks;
(b) beach location parameters;
(c) reconnaissance methods and techniques;
(d) survey equipment; and
(e) reports and returns;

(3) explain:

(a) the requirement for beach surveys;
(b) tasks; and
(c) beach location parameters;

(4) describe the following beach reconnaissance, explaining the advantages / disadvantages, personnel and equipment requirements and procedures for each:

(a) parallel administrative;
(b) perpendicular administrative;
(c) inflatable boat system (IBS);
(d) night combat; and
(e) day combat; and

(5) explain use of tidal board;

(a) post-reconnaissance questionnaire;
(b) swimmer’s slate (day combat);
(c) perpendicular administrative, IBS and submerged; and
(d) parallel administrative and night combat.

EC 404 .4 Explain CCDA operations

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom.

ARRANGEMENTS AND PROCEDURES

2. The officer will complete a written test on the operators of the CCDA IAW classroom lecture and instructor demonstration.

3. The officer will complete a written test on terminology used in Beach Survey Operations IAW classroom lecture.

4. Assistance and references are denied.

5. Pass mark is 80%.

6. Time required - 1 period for CCDA and 1 period for Beach Survey Operations.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of the operation of the CCDA and of terminology used in Beach Survey Operations.

2. Assistance and references are denied.

3. Pass mark is 80%.

4. Time required - 1 period for CCDA and 1 period for Beach Survey Operations.

PART III - ENABLING CHECK NOTES

1. Test Plan: CCDA

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE (%)
CCDA General	applications	10
	components	10
	limitations / capabilities /capacities	
	breathing system	
	rules and regs	10
Function and Operation	buoyancy control valve	
	counter lung relief valve	5
	back-weight pocket release mechanism	5
	flow selector and by-pass valve assembly	10
	reserve valve	5
	absolute pressure regulator	5
	scrubber canister	5

2. Test Plan: Beach Survey

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE (%)
Survey terminology	types of breakers	10
	beach terminology	50
	tide terminology	40

PO 404 Supervise Self-Contained Breathing Apparatus (SCBA) Operations.

EO 404 .5

1. Performance Conduct CCDA Operations.

2. Conditions.

   a. Given:

      (1) test and calibration equipment;
      (2) gas booster pump;
      (3) CCDA;
      (4) MCM beach survey equipment;
      (5) dive team;
      (6) supervision for safety;
      (7) emergency scenario;
      (8) support platform and RCC;
      (9) technical support; and
      (10) references.

   b. Denied: assistance.

   c. Environmental: in existing weather conditions, day or night.

3. Standard. The officer shall conduct CCDA operations IAW refs to include:

   a. carry out the following procedures:

      (1) dressing in / out and pre / post dive procedures;
      (2) minefield entry / exit;
      (3) conduct CCDA dive IAW MCM SOPs;
      (4) carry out emergency procedures by the scenario; and
      (5) beach survey operations;

   b. accumulate the following minimum dive times:

      (1) 26 hours - oxygen: at least three dives using oxygen shall be a minimum of one hour duration;
      (2) 34 hours - mixed gas; completed a minimum of one dive to within 20 fsw of the maximum operational limit for each of the gas mixtures.

4. Time 130 x 50 min periods

5. Method media

   a. Lecture demonstration - 3 periods
   b. practical - 92 periods; and c. EC - 35 periods.

6. References. A9, A31, A33 Book 1.

7. Training Aids. Equipment IAW condition statement.

8. Test Details. Practical test. Pass mark SAT.

9. Remarks.

   a. Ensure that each student fully understands all the emergency procedures prior to making any dives in

CCDA;

b. Ditching drill should be carried out in FDU(A) training tank and an RCC will be on site and on standby.
c. This EC is to be completed in conjunction with EC 404 .6 ; and
d. any dives deeper than 150 fsw will require an RCC on site.

10. Teaching Points. Explain dive and post-dive procedures IAW references to include:

a. explain the procedure for dressing;
b. explain the MCM dive SOPs to include:

(1) MCM equipment requirements / preparation including:

(a) boat and engine;
(b) hand-held GPS / vectoring;
(c) hand-held sonar;
(d) entry / exit / master reference buoys;
(e) search line;
(f) lazy shot procedure; and
(g) MCM signals;

(2) minefield discipline to include:

(a) entry / exit procedures;
(b) laying entry / exit buoy / MRB;
(c) silent water entry / leak check / exit;
(d) MCM dive SOPs;
(e) operation of the by-pass;
(f) buoyancy control;
(g) flushing through the counter lung;
(h) surfacing temporarily;
(i) CCDA rigged for oxygen breathing; and
(j) condensate in the breathing circuit;

c. explain the emergency SOPs to include:

(1) trouble drill;
(2) emergency ascent;
(3) buddy drill;
(4) flask exhaustion;
(5) emergency use of by-pass;
(6) oxygen deficiency;
(7) C02 poisoning;
(8) Narcosis; and
(9) caustic cocktail;

d. conduct beach survey operations to include:

(1) beach party / dive team;
(2) dive / survey equipment; and
(3) area to be surveyed;

e. explain the standby diver procedures; and
f. explain the action in the event of a diving accident in CCDA.

EC 404 .5

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. CCDA.

2. Maintenance equipment.

3. Support personnel.

4. Diving support vessel.

ARRANGEMENTS AND PROCEDURES

5. The officer will perform dives using CCDA.

6. Assistance is denied; references are allowed

7. Supervision will be supplied for safety.

8. Pass mark is a grading of SAT.

9. Time required - 35 periods.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge to perform dives using CCDA.

2. Assistance is denied; references are allowed.

3. Supervision will be supplied for safety.

4. Pass mark is a grading of SAT.

5. Time required - 35 periods.

PART III - ENABLING CHECK NOTES

1. Not used.

PART IV - ENABLING CHECK LIST

EC 404 .5

Student: Date: SAT/UNSAT:

CHECKLIST	SAT	UNSAT	COMMENTS
1. Carried out pre / post dive procedures.			
2. Conducted pre-dive drill correctly.			
3. Water entry.			
4. Carry out dive SOP.			
5. Successfully completed one ditching drill.			
6. Successfully carried out buddy drill.			
7. Completed 20 hours of diving on oxygen.			
8. Completed 6 hours of diving on Nitrox.			
9. Completed 8 compass swims.			
10. Completed a minimum of one dive to within 20 fsw of the max operational depth for each of the gas mixtures.			
11. Observed all safety precautions. *			

Notes:

a. * Denotes critical check.

b. Trainee must score a SAT on all critical checks to achieve an overall grading of SAT.

c. A scoring of UNSAT on two non-critical checks will be reviewed by the Senior Instructor / Trg CPO to determine the effect on the overall grade.

PO 404 Supervise Self-Contained Breathing Apparatus (SCBA) Operations.

EO 404 .6

1. Performance. Supervise CCDA Operations.

2. Conditions.

   a. Given:

      (1) equipped dive team;
      (2) diving emergency scenario;
      (3) supervision for safety;
      (4) tasking; and
      (5) references.

   b. Denied: assistance.

   c. Environmental: in existing weather conditions, day or night.

3. Standard. The officer shall supervise CCDA OPS IAW references to include:

   a. supervise a minimum of:

      (1) 1 dive using oxygen; and
      (2) 3 dives using mixture;

   b. ensure the breathing gas meets specifications;

   c. ensure the following MCM procedures are carried out IAW references:

      (1) equipment preparations;
      (2) pre-dive checks;
      (3) minefield discipline;
      (4) MCM dive procedures; and
      (5) post-dive routines;

   d. calculate EAD;

   e. complete repetitive dive worksheets;

   f. direct a response to an emergency scenario IAW references ; and

   g. ensure beach survey procedures are carried out.

4. Time. 10 x 50-minute periods.

5. Method/ Media demonstration / practical

6. References. A3, A33 (book 1).

7. Training Aids. Not used.

8. Test Details. Pass mark is SAT on checklist

9. Remarks.

a. Each student should supervise a minimum of one dive using oxygen and one dive using mixture, prior to carrying out the EC;

b. Each student will still be supervised by the course Instructor.

10. Teaching Points.

a. No additional teaching points are required for this EO.

EC 404.6

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. CCDA.

2. Maintenance equipment.

3. Support personnel.

4. Diving support vessel.

ARRANGEMENTS AND PROCEDURES

5. The officer will supervise dives using CCDA with the instructor providing supervision for safety.

6. Assistance is denied.

7. Pass mark is a grading of SAT on the ECL.

8. Time required - 10 periods.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will demonstrate your ability to supervise dives using CCDA.

2. Assistance is denied.

3. Pass mark is a grading of SAT on the ECL.

4. Time required - 10 periods.

PART III - ENABLING CHECK NOTES Nil

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PART IV - ENABLING CHECK LIST

Student: Date: SAT: UNSAT:

CHECKLIST	SAT	UNSAT	COMMENTS
1. Supervised a minimum of two (2) 02 dives.			
2. Supervised a minimum of two (2) dives using a mixture.			
3. Ensured breathing media meets specifications.			
4. Equipment preparations carried out.			
5. Pre-dive checks done.			
6. Dive procedure carried out.			
7. Post-dive routines carried out.			
8. Calculated EAD.			
9. Completed repetitive dive work sheet.			
10. Directed a response to an emergency scenario.			
11. Observed all safety precautions. *			

Notes:

a. * Denotes critical check.

b. Trainee must score a SAT on all critical checks to achieve an overall grading of SAT.

c. A scoring of UNSAT on two non-critical checks will be reviewed by the Senior Instructor / Trg CPO to determine the effect on the overall grade.

PO 404 Supervise Self-Contained Breathing Apparatus (SCBA) Operations.

EO 404 .7

1. Performance. Explain CUMA Operations.

2. Conditions.

   a. Given:

      (1) classroom;
      (2) diving / MCM equipment; and
      (3) technical support.

   b. Denied:

      (1) references; and
      (2) assistance.

   c. Environmental: n/a.

3. Standard. The trainee shall explain / describe:

   a. CUMA rules and regulations;
   b. CUMA safety regulations;
   c. the function and operation of CUMA components;
   d. auxiliary equipment; and
   e. methods of storage and handling of CUMA.

4. Time. 2 x 50-minute periods.

5. Method/Media. lecture.

6. References. A8, A30, A31, A33 book 2, A47, C6, C7.

7. Training Aids.

   a. CUMA components;
   b. OHP;
   c. handouts: and

8. Test details Written exam - Pass mark is 80 %

9. Remarks. None.

10. Teaching Points.

    a. Rules and regulations IAW ref Chap 1 sec. 2 to include:

       (1) diving responsibilities art 1201; (2) standby diver art 1202; (3) concept of operations art 1203; and (4) dive planning art 1204.

    b. Safety regulations IAW ref book 2, Chap 1 Sec. 3 and 5 to include:

       (1) minimum personnel required art 1301;
       (2) depth limitations art 1302;
       (3) training and proficiency of divers art 1303;
       (4) diving equipment art 1304;

(5) use of bailout art 1305;
(6) MCM diving art 1306;
(7) operating in vicinity of live ordnance art 1307; and
(8) action in the event of a diving accident.

c. CUMA components IAW ref Book 2, Chap 4 Sec. 1 and 2 to include:

(1) description art 4101;
(2) depth limit art 4102;
(3) applications art 4103;
(4) design concept art 4201;
(5) harness, counter-lung and counter weights art 4202;
(6) back pack art 4203;
(7) C02 canister art 4204;
(8) hoses, mouth piece and mask art 4205;
(9) gas supply system art 4206;
(10) PP02 diagnostic assembly art 4207;
(11) submersible gauges art 2408;
(12) onboard bailout art 4209; and
(13) components art 4210.

d. Auxiliary equipment IAW ref Chap 4 Sec. 2 to include:

(1) additional supporting equipment art 4211; and
(2) volumes, weights and capacities art 4212.

e. Handling and storage of equipment IAW ref Chap 5 Sec. 1 to include:

(1) handling the apparatus art 5110;
(2) storage and care of CUMA art 5111 : and
(3) magnetic cleanliness art 5112

EC 404 .7

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom.

ARRANGEMENTS AND PROCEDURES

2. The trainee will complete a written exam on the operation of CUMA IAW classroom lecture and instructor demonstration.

3. Assistance and references are denied.

4. Time allowed is 1 period.

5. Pass mark is 80%.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of the operation and safe usage of the CUMA set.

2. Assistance and references are denied.

3. Pass mark is 80%.

4. Time allotted is 1 period.

PART III - ENABLING CHECK NOTES

1. Test plan:

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE (%)
CUMA operations	Rules / regulations	40
	Safety regulations	40
	CUMA components	10
	Auxiliary equipment	5

PO 404 Supervise Self-Contained Breathing Apparatus (SCBA) Operations.

EO 404 .8

1. Performance. Apply Mixed Gas Breathing Theory.

2. Conditions.

   a. Given:

      (1) decompression tables;
      (2) references (tables and formulas as required); and
      (3) classroom.

   b. Denied: assistance.

   c. Environmental: n/a.

3. Standard. The trainee shall:

   a. explain CUMA theory;
   b. describe CUMA diving emergencies;
   c. describe rebreather diving emergencies; and
   d. calculate underwater endurance and decompression.

4. Time. 6 x 50-minute periods.

5. Method/Media.

   a. lecture - 3 periods;
   b. EC (written) - 2 periods; and
   c. review EC - 1 period.

6. References. A33.

7. Training Aids.

   a. whiteboard;
   b. multi media systems;
   c. handout “Rebreather diving emergencies” FDU(A) 4500-2 12 Mar 92

8. Test Details Written exam - Pass mark is 80 %

9. Remarks. Have students read handout prior to lecture.

10. Teaching Points.

    a. CUMA theory IAW ref book2 Chap 1 Sect 1 to include:

       (1) introduction art 1101;
       (2) CUMA gas mixing art 1102;
       (3) PP02 in breathing loop art 1103;
       (4) thermal problems art 1105;
       (5) speech distortion art 1106; and
       (6) formulas for gas usage rates aft 1107.

b. CUMA diving emergencies IAW ref Book 2 Chap 2 Sec. 1 and 2 to include:

(1) introduction art 2101; and
(2) blow-up art 2201.

c. Rebreather diving emergencies IAW ref Book 1 Chap 2 Sec. 2 to include:

(1) oxygen deficiency art 2201;
(2) CNS toxicity art 2202;
(3) C02 poisoning art 2203; and
(4) caustic cocktail art 2205.

d. Underwater endurance IAW ref Book 2 Chap. 3 to include:

(1) modes of operation art 3101;
(2) gas endurance art 3102;
(3) canister endurance art 3103;
(4) endurance / operating limits art 3104;
(5) 02 breathing limits art 3105;
(6) available diluent gas in the flask fig 3-1;
(7) diluent gas usage fig 3-2; and
(8) operational data fig 3-3.

EC404.7 and 404.8 Explain CUMA Operations and Apply Mixed Gas Breathing Theory.

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom.

ARRANGEMENTS AND PROCEDURES

2. The trainee will complete a written multiple-choice test on mixed gas for CUMA

3. Assistance is denied, formulas and tables will be provided.

4. Pass mark - 80%.

5. Time required - 2 periods.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of the mixed gas theory for CUMA.

2. You are denied assistance; formulas and tables will be provided as necessary.

3. Pass mark - 80% on a multiple-choice exam.

4. Time allowed - 2 periods.

PART III - ENABLING CHECK NOTES

1. Test Plan:

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE
CUMA mixed gas breathing theory	theory	
	diving emergencies	
	rebreather emergencies	
	underwater endurance	

PART IV - ENABLING CHECKLIST N/A

PO 404 Supervise Self-Contained Breathing Apparatus (SCBA) Operations.

EO 404 .9

1. Performance. Prepare CUMA for Diving.

2. Conditions.

   a. Given:

      (1) CUMA and associated equipment;
      (2) calibration panel;
      (3) tool kit;
      (4) Haskel booster pump;
      (5) references;
      (6) supervision for safety;
      (7) helium, oxygen and 30 / 70 02He gases; and
      (8) soda sorb.

   b. Denied: assistance.

   c. Environmental: an 02 clean workspace.

3. Standard. The trainee shall, IAW references:

   a. prepare COMA for diving using pre-dive checklists;
   b. carry out following pre-dive preparations to include:

      (1) operate Haskel booster pump; and
      (2) charge CUMA flasks;

   c. complete CUMA maintenance log; and
   d. apply all safety precautions.

4. Time. 11 x 50-minute periods.

5. Method/Media.

   a. lecture - 2 periods; and
   b. demonstration and practical - 9 periods.

6. References. A31, A33, A47.

7. Training Aids. Equipment as per condition statements.

8. Test Details. Written test. Pass mark - 80%.

9. Remarks. Not used.

10. Teachings Points.

a. Explain and demonstrate flask charging procedures IAW ref A47 and SOPs.

b. Explain and demonstrate pre and post dive SOPs IAW ref A33 Book 2 Chap 5 SEc. 1 to include:

(1) pre- and post-dive routines;
(2) pre-dive assembly;
(3) pre-dive checklist;
(4) charging absorbent canisters;
(5) install mask / hose assembly;
(6) gas supply and gauge check;
(7) gas circuit calibration check;
(8) leak test;
(9) CUMA pre-dive tag;
(10) post-dive routine; and
(11) set preparation between dives.

c. Explain and demonstrate maintenance and calibration procedures IAW A33 Book 2 Chap 6 Sec. 1 to include:

(1) conditions for calibration and maintenance;
(2) calibration check;
(3) leak test;
(4) by-pass and regulators CUMA test #4;
(5) oxygen system CUMA test #5;
(6) diluent system CUMA test #6;
(7) PP02 diagnostic assembly COMA test #7;
(8) fuel cell CUMA test #8;
(9) CRV and BCV CUMA test #9; and
(10) bailout assembly CUMA test #10.

d. Complete maintenance logs IAW A33 Book 2 Chap 6 sect. 1.

PO 404 Supervise Self-Contained Breathing Apparatus (SCBA) Operations.

EO 404.10

1. Performance. Conduct CUMA Operations.

2. Conditions.

   a. Given:

      (1) test and calibration equipment;
      (2) HE and 02 gases;
      (3) CUMA and associated equipment;
      (4) dive team and support equipment;
      (5) supervision;
      (6) simulated emergency scenario; and
      (7) references.

   b. Denied: assistance.

   c. Environmental: existing weather conditions, day or night.

3. Standard. The trainee shall, IAW references:

   a. carry out the following procedures:

      (1) dressing in;
      (2) systems check;
      (3) in water check;
      (4) 02 flushing drill;
      (5) surface temporarily;
      (6) surfacing drill;
      (7) recover into zodiac drill;
      (8) trouble drill green light;
      (9) trouble drill red light;
      (10) emergency ascent;
      (11) buddy drill; and
      (12) dressing out drill;

   b. accumulate a minimum of 4 hours water time (time does not include RCC decompression);

   c. dive to a minimum of 2/3 of the maximum operational depth;

   d. respond appropriately to emergency scenario (simulated):

   e. be able to use CUMA tables;

   f. demonstrate CUMA signals;

   g. employ shot line procedures; and

   h. apply all safety precautions.

4. Time. 86 x 50-minute periods.

5. Method/Media.

a. lecture; - 3 periods;
b. demo and practical - 68 periods;
c. EC (practical) - 14 periods; and
d. EC review - 1 period.

6. References. A33 book 2.

7. Training Aids. Equipment as per condition statement.

8. Test Details. This EO will be tested in conjunction with EO 404 .11

9. Remarks.

a. All emergency procedures will be carried out in a test tank or swimming pool.

b. When conducting the ditching drill, an RCC will be on site and in a state of immediate readiness.

c. co-ordination is required with the Operations, Boats and Repair Department for the use of a dive platform, RCC< COMA equipment, gases, CUMA lab and charging facilities.

d. Actual decompression is required to dive CUMA to a minimum of 2/3 of the maximum operational depth which is 180' of 270'.

e. A RCC is required on site for any dives deeper than 150 FSW.

10. Teaching Points.

a. Explain and demonstrate SOPs for diving drills IAW reference book 2 Chap 5 Sect 2 to include:

(1) dressing in;
(2) systems check;
(3) in-water check;
(4) dive;
(5) flushing through the counter lung;
(6) surfacing temporarily
(7) surfacing drill;
(8) recovery zodiac;
(9) procedure for standby diver;
(10) ascent;
(11) dressing out; and
(12) treatment of a flooded set.

b. Explain and demonstrate emergency procedures IAW reference, Book 2 Chap 5 Sec. 2 to include:

(1) trouble drill;
(2) red light drill;
(3) emergency ascent;
(4) ditching drill;
(5) buddy drill;
(6) flask exhaustion; and
(7) emergency use of by-pass;

c. Utilize CUMA decompression procedures IAW reference, Book 2, Chap 3, Sec. 2 to include:

(1) definition;
(2) operational procedures; and
(3) dive abort direction and combined bottom time / maximum depth option.

d. Utilize decompression table procedure for CUMA diving IAW reference, Book 2, Chap 3, Sec. 3 to include:

(1) gases;
(2) helium-oxygen gas mixture;
(3) oxygen;
(4) exceptional exposure;
(5) travel rates;
(6) stop and travel times;
(7) 30 FSW 02 decompression stop;
(8) delays;
(9) oxygen toxicity at 02 stop;
(10) loss of 02;
(11) Sur-D O2 RCC time;
(12) Surface interval violation of 7 mins, surface interval and omitted decompression;
(13) flying after diving;
(14) decompression stress during surface interval; and
(15) decompression.

e. Demonstrate CUMA signals IAW reference, Book 2, Chap 2, Sec. 4 to include:

(1) diver communication;
(2) supplementary signals; and
(3) rehearsal.

f. Employ shot line procedure IAW reference, Book 2, Chap. 1, Sec. 4 to include:

(1) MCM dry suit and suit inflation assembly;
(2) support craft;
(3) diving shots and lines;
(4) pre-dive depth check; and
(5) general arrangement.

EC 404 .9 and 404 .10.

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. CUMA

2. Maintenance equipment.

3. Qualified support personnel.

4. Diving support vessel.

5. Oxygen and He02 gas.

ARRANGEMENTS AND PROCEDURES

6. The trainee will prepare and dive CUMA.

7. Assistance is denied, references are allowed.

8. Pass mark is a grading of SAT on the checklist.

9. Time required - approximately 4 periods per student which includes preparation in time and travel to the dive site.

10. The instructor will provide direct supervision at all times.

11. Any safety violations will constitute a failure.

12. You will have to co-ordinate with the Boats Department and the MCM team for use of the dive platform, RCC and CUMA equipment. You will also have to co-ordinate with the Repair Department for gases to be used, CUMA Lab, and charging facilities.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will demonstrate your ability to prepare and dive CUMA.

2. You are denied assistance.

3. Reference B-GG-380 Vol 4, Book 2 is allowed.

4. Pass mark is a grading of SAT on the checklist.

PART IV - ENABLING CHECKLIST

EOs 404 .9 and 404 .10.

Performance: Prepare and Dive CUMA.

Student: Date: SAT: UNSAT:

CHECKLIST	Sat	Unsat	COMMENTS
*1. Prepare CUMA set:			
a, pre-dive checks			
b. post-dive checks			
2. Operate gas booster pump to charge CUMA flasks			
3. Complete CUMA maintenance log			
4. Accumulate 4 hours in-water time			
*5. Dive to minimum of 180 FSW			
*6. Respond appropriately to a simulated diving emergency			
7. Demonstrate CUMA signals			
8. Employ shot-line procedures			
*9. Carry out diving drills:			
a. dressing in			
b, systems check			
c. in-water check			
d. flushing through C/L			
e. surface temporarily			
f. surfacing drill			
g. recover into zodiac			
h. dressing out			
*10. Demonstrate emergency procedures:			
a. trouble drill			
b. red light drill			
c. buddy drill			
d. ditching drill			
e. emergency ascent			
*11. Conduct dive IAW SOPs			
*12 Apply all safety precautions			

a. * Denotes critical check.

b. Trainee must score a SAT on all critical checks to achieve an overall grading of SAT.

c. A scoring of UNSAT on two non-critical checks will be reviewed by the Senior Instructor / Trg CPO to determine the effect on the overall grade.

EO 404.11

1. Performance. Supervise CUMA Operations.

2. Conditions.

   a. Given:

      (1) fully equipped dive team;
      (2) diving emergency scenario;
      (3) supervision for safety;
      (4) task; and
      (5) references.

   b. Denied: assistance.

   c. Environmental: In existing weather condition, day and night.

3. Standard. The officer shall supervise CUMA ops IAW references to include:

   a. Ensure breathing gas meets specification;

   b. Ensure the following MCM procedures are carried out IAW references:

      (1) equipment preparation;
      (2) pre-dive checks;
      (3) mine field discipline;
      (4) MCM dive procedure; and
      (5) post-dive routines;

   c. calculate EAD;

   d. direct a response to an emergency scenario IAW references; and

   e. ensure beach survey procedures are carried out.

4. Time. 10 x 50-minute periods.

5. Method/Media.

6. References A33 BOOK 2, C6, C7

7. Training Aids. None

8. Test Details. Practical test Pass mark SAT

9. Remarks. Each student will be supervised by the course instructor.

10. Teaching Points. All teaching points have been covered in previous EOs.

EC 404.11

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1 . CUMA sets.

2. Maintenance equipment and facilities for CUMA.

3. Oxygen and Helium for diving.

4. Qualified support personnel.

5. Diving support vessel.

ARRANGEMENTS AND PROCEDURES

6. The officer will supervise dives using CUMA with instructor providing supervision for safety.

7. One (1) 342 Clearance Diver Technician proficient in CUMA maintenance pro-cedures.

8. If the instructor feels that safety has been or will be violated, he / she shall assume control of the dive team.

9. Pass mark is a grading of SAT on the checklist.

10. Assistance is denied.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will demonstrate your ability to supervise dives using CUMA.

2. Assistance is denied.

3. Pass mark is SAT on the checklist.

4. If the instructor feels that safety has been or will be violated, he / she shall assume control of the dive team.

PART III - ENABLING CHECK NOTES None.

PART IV - ENABLING CHECK LIST

Student: Date: SAT: UNSAT:

CHECKLIST	SAT	UNSAT	COMMENTS
1. Supervise two in-water decompression dives*			
2. Supervise one surface decompression dive*			
3. Ensure breathing gas meets specification			
4. Equipment preparation carried out			
5. Pre-dive checks complete			
6. Dive procedures carried out			
7. Post dive routines carried out			
8. Calculate EAD*			
9. Direct a response to an emergency scenario			
10. Observed all safety precautions*			
11. Completed repetitive dive			

Notes:

a. * Denotes critical check.

b. Trainee must score a SAT on all critical checks to achieve an overall grading of SAT.

c. A scoring of UNSAT on two non-critical checks will be reviewed by the Senior Instructor / Trg CPO to determine the effect on the overall grade.

PC 404

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. SCBA equipment.

2. Technical support.

3. Diving support vessel.

ARRANGEMENTS AND PROCEDURES

4. The trainee will perform / supervise dives in open circuit (CABA), and semi-closed / closed (CCDA and CUMA) breathing apparatus.

5. References are allowed in some practical ECs but no references are allowed for written exams except for dive tables.

6. Pass mark is a grading of SAT on the PC checklist.

7. The trainee is to be informed when an EC checklist is being used as a PC.

8. For assessment purposes, to satisfy the standard to dive to the maximum oper-ational depth: dives in CCDA to be conducted to within 6 MSW (20 FSW); and CABA and CUMA dives to a minimum of 2/3 operational depth.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This PC will demonstrate your ability to dive and supervise CABA, CCDA and dives.

2. References are allowed in some ECs, assistance is denied.

3. Pass mark is a grading of SAT on the PC checklist.

PART III - PERFORMANCE CHECK NOTES

1. Not used.

PART IV - PERFORMANCE CHECK LIST

PO 404

Student: Date: SAT: UNSAT:

CHECKLIST	YES	NO	COMMENTS
1. Explain diving rules and regulations			
2. Perform dives using CABA			
3. Apply diving physics			
4. Apply 02 and mixed gas breathing theory			
5. Explain the operation of CCDA			
6. Carry out CCDA pre-dive tests			
7. Charge CCDA gas cylinders			
8. Perform dives using CCDA.			
9. Supervise CCDA dive			
10. Operate hand-held sensors			
11. Explain diving physiology and medicine			
12. Explain operation of CUMA and apply mixed gas breathing theory			
13. Prepare CUMA for diving and perform dives using CUMA			
14. Supervise CUMA dive			

Notes:

a. * Denotes critical check.

b. Trainee must score a SAT on all critical checks to achieve an overall grading of SAT.

c. A scoring of UNSAT on two non-critical checks will be reviewed by the Senior Instructor / Trg CPO to determine the effect on the overall grade.

PO 405 Supervise Demolition Operations.

EO 405 .1

1. Performance. Explain Rules and Regulations for Demolition Tasks.

2. Conditions.

   a. Given: n/a

   b. Denied:

      (1) assistance; and
      (2) references.

   c. Environmental: n/a.

3. Standard. The officer shall explain:

   a. field storage and transportation;

   b. safety distances;

   c. preparation of charges;

   d. duties of a Range Safety Officer (RSO); and

   e. duties of a sentry.

4. Time. 3 x 50-minute periods.

5. Method/Media. lecture.

6. References. A7, A39.

7. Training Aids. Not used.

8. Test Details. Written exam. Pass mark - 80%.

9. Remarks. None.

10. Teaching Points

   a. Explain rules governing storage and transportation, including;

      (1) conditions of storage;
      (2) fire precautions; and
      (3) transportation regulations concerning vehicles.

   b. Explain safety distances, to include:

      (1) timber cutting;
      (2) concrete beams and pillars;
      (3) metal targets;
      (4) cratering; and

(5) Bangalore torpedoes and shaped charges.

c. Explain preparation of charges, to include:

(1) rules governing preparation;
(2) precautions for electric initiation;
(3) safety procedures before firing;
(4) procedure on firing; and
(5) procedure after firing.

d. Explain duties of an RSO, to include:

(1) responsibilities of an RSO;
(2) duties prior to conducting a range practice;
(3) duties during a range practice; and
(4) duties after a range practice.

e. Explain duties of a sentry, to include:

(1) signals to be used;
(2) authorized entry;
(3) start and stop times and signals; and
(4) reporting to RSO.

EC 405 .1

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom.

ARRANGEMENTS AND PROCEDURES

2. The officer will complete a written test on rules and regulations for field demolition tasks IAW classroom lecture.

3. Assistance and references are denied.

4. Pass mark - 80%.

5. Time required - 1 period.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of rules and regulations for field demolition tasks.

2. You are denied assistance and references.

3. Pass mark - 80% on a written test.

4. Time allowed - 1 period.

PART III - ENABLING CHECK NOTES

1.

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE (%)
Storage and transportation rules	conditions of storage	7
	fire precautions	10
	vehicle transport regulations	10
Safety distances	timber cutting	3
	concrete beams / pillars	3
	metal targets	3
	cratering and Bangalore torpedoes/ shaped charges	3
Charge preparation	rules and precautions governing preparation / electric initiation	15
	safety procedures	10
	firing procedures	10
Duties of the sentry	signals used and authorized entry	3
	start / stop times and reporting to RSO	3

PO 405 Supervise Demolition Operations.

EO 405 .2

1. Performance. Calculate Explosive Requirements for Demolition Tasks.

2. Conditions.

   a. Given:

      (1) references; and
      (2) calculators.

   b. Denied: assistance.

   c. Environmental: n/a.

3. Standard. The officer shall calculate explosive requirements for demolition tasks to include:

   a. relative effect factor;
   b. Hasty Tables;
   c. RECCEE report;
   d. ice demolitions; and
   e. underwater demolitions.

4. Time. 5 x 50-minute periods.

5. Method/Media. lecture / demonstration / performance - 4 periods; and b. EC - 1 period.

6. References. A7, A8, A39.

7. Training Aids.

   a. whiteboard;
   b. OHP; and
   c. handouts (tables).

8. Test Details Written exam. Pass mark is 80 %

9. Remarks This phase is conducted completely in classroom prior to field indoctrination

10. Teaching Points.

    a. Explain relative effect factors and conversions for various explosives.

    b. Explain Hasty Tables as pertaining to:

       (1) steel cutting;
       (2) "I" beam cutting;
       (3) timber cutting;
       (4) concrete breaching; and
       (5) shaped charges and cratering.

c. Explain purpose and properly complete RECCEE report as pertaining to:

(1) bridges / piers; and
(2) standing structures.

d. Explain procedures for ice demolitions as pertaining to:

(1) calculations; and
(2) placement.

e. Explain the difference between surface and underwater demolitions as pertaining to:

(1) tamping;
(2) speed of safety fuse;
(3) safety distances;
(4) preferred method of initiation; and
(5) personnel required.

EC 405.2

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom.

2. Calculators.

ARRANGEMENTS AND PROCEDURES

3. The officer will complete a written test on placement, amount and type of explosives required for demolition tasks IAW classroom lecture

4. Assistance is denied,

5. Pass mark - 80%.

6. Time required - 1 period.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of placement, amount and type of explosives required for demolition tasks.

2. You are denied assistance Hasty Tables and calculators are to be provided.

3. Pass mark - 80% on a written test.

4. Time allowed - 1 period.

PART III - ENABLING CHECK NOTES

1. Test Plan:

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE (%))
Relative effect for various explosives	factors and conversions	15
Hasty tables	steel cutting	10
	“I” beam cutting	10
	timber cutting	10
	concrete breaching	10
	shaped charges/ cratering	10
RECCEE report	bridges / piers	10
	standing structures	10
Ice demolition procedures	calculators	10
	placement	5

PO 405 Supervise Demolition Operations.

EO 405 .3

1. Performance. Conduct Surface Demolition Task.

2. Conditions.

   a. Given:

      (1) explosives;
      (2) test equipment;
      (3) firing devices electric / non-electric;
      (4) obstacles;
      (5) references;
      (6) range site; and
      (7) supervision for safety.

   b. Denied: assistance.

   c. Environmental: within environmental and local limits contained in Range Standing Orders and ref A7.

3. Standard. The officer shall:

   a. determine the min. amount/type of explosives needed for cutting obstacles;
   b. determine safe distance for personnel;
   c. complete proper firing circuits;
   d. comply with misfire procedures;
   e. fire surface demolitions; and
   f. inspect the demolition site and complete reports.

4. Time. 3 x 50-minute periods.

5. Method/Media.

   a. lecture; and
   b. practical.

6. References. A7, A10, A12, A39.

7. Training aids

   a. whiteboard;
   b. OHP; and
   c. handouts.

8. Test Details. Written test. Pass mark - 80%.

9. Remarks. Not used.

10. Teaching Points. All teaching points have been covered in previous EOs.

PO 405 Supervise Demolition Operations.

EO 405 .4

1. Performance. Conduct Underwater Demolition Task.

2. Conditions.

   a. Given:

      (1) explosives;
      (2) test equipment;
      (3) firing devices electric / non-electric;
      (4) obstacles;
      (5) references;
      (6) range site;
      (7) dive team; and
      (8) supervision for safety.

   b. Denied: assistance.

   c. Environmental: within environmental and local limits contained in Range Standing orders and reference A7.

3. Standard. The officer shall:

   a. determine the min amount/type of explosives required for cutting obstacles;
   b. determine safe distance for personnel;
   c. complete proper firing circuits;
   d. comply with misfire procedures;
   e. fire underwater demolitions; and
   f. inspect the demolition site and complete reports.

4. Time. 1 x 50-minute periods.

5. Method/Media. lecture; and b. practical.

6. References. A2, A3, A7, A8, A10, A12, A39.

7. Training Aids

   a. whiteboard;
   b. multi media system; and
   c. handouts

8. Test Details. Written test. Pass mark - 80%.

9. Remarks. Not used.

10. Teaching Points. All teaching points have been covered in previous EOs

PO 405 Supervise Demolition Operations.

EO 405 .5

1. Performance. Supervise Demolition Tasks.

2. Conditions.

a. Given:

(1) demolition team;
(2) explosives;
(3) ordnance for disposal;
(4) obstacles for demolition;
(5) demolition kit;
(6) transport;
(7) OPORD**;** and
(8) Range Safety Orders; and (9) refs / SOPs.

b. Denied: assistance.

c. Environmental: within environmental and local limits contained within Range Standing Orders and ref A7.

3. Standard. The officer shall:

a. supervise at least one of the following:

(1) cutting steel plate;
(2) cutting steel "I" beam;
(3) cutting timber (minimum 6" thickness);
(4) cutting steel wire rope;
(5) breaching concrete; and
(6) one underwater demolition task;

b. supervise the disposal of any suitable ordnance; and

c. ensure that the demolition is carried out IAW OPORD**.**

4. Time. 14 x 50-minute periods.

5. Method/Media. practical.

6. References A7, A10, A39, A56.

7. Training Aids. Not used.

8, Test Details. Practical assessment.

9. Remarks. This EO is based on knowledge and skills gained in previous EOs.

10. Teaching Points. Not used.

EC 405.5

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Range.

2. Transport.

3. Explosives.

4. Material for demolition.

ARRANGEMENTS AND PROCEDURES

5. The officer will supervise a surface demolition operation. Instructor will supply supervision for safety.

6. Assistance is denied.

7. Pass mark is a grade of SAT on checklist.

8. Time required - 14 periods.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will demonstrate your ability to supervise a surface demolitions operation.

2. You are denied assistance.

3. Pass mark is a grading of SAT on the checklist.

4. Time allowed - 14 periods.

PART III - ENABLING CHECK NOTES Not used.

PART IV - ENABLING CHECK LIST

EO 405 .5

Performance: Supervise Demolition Task.

Student:_____ Date:_____ SAT:_____UNSAT:_____

CHECKLIST	Sat	Unsat	COMMENTS
1. Has range standing orders in hand.			
2. Ensure officer has read:			
a. Ref A10			
b. C-09-008-002 / FP-000 Destruction of dud and misfired ammo			
3. Ensure all equipment required is on hand			
4. Observes all safety precautions			
5. Confirms method of disposal			
a. detonation (for electric (E) or non-electric (NE))			
b. vent and burn			
c. burn			
6. Positioned stores in safe area for destruction.*			
7. Moves stores in destruction point separately and safely			
8. Prepares pit as per section 1 of A10			
9. Clears area			
10. Connections for blasting caps prepared (E)			
11. Primes initiating charge			
12. Counts all personnel			
13. Checks continuity (E)			
14. Connect to blasting machine (E)			
15. Area check			
16. Has possession of handle to blasting machine			
17. Fires charge IAW A10, sect 3, para 14			
18. Carries out misfire procedures IAW A10, Sec 3, para 16			
19. Carries out all clear procedure			

FOR NON-ELECTRIC FIRING			
CHECKLIST	Sat	Unsat	COMMENTS
22. Positions stores for destruction in safe area IAW A10			
23. Moves stores safely			
24. Times safety fuse IAW Section 4, para 7 of A10			
25. Places charge properly			
26. Clears area properly			
27. Crimps blasting cap properly			
28. Primed initiating charge			
29. Area check			
30. Counts personnel			
31. All personnel in safe area			
32. Observation of the detonation			
33. All clear procedure			
34. Misfire procedures IAW Sec 4 para 21 of A10*			

Notes:

a. * Denotes critical check.

b. Trainee must score a SAT on all critical checks to achieve an overall grading of SAT.

c. A scoring of UNSAT on two non-critical checks will be reviewed by the Senior Instructor / Trg CPO to determine the effect on the overall grade.

PO 406 Supervise Surface Supplied Breathing Apparatus (SSBA) Operations.

EO 406 .1

1. Performance. Explain SSBA Rules and Regulations.

2. Conditions.

   a. Given: nil.

   b. Denied: assistance and references.

   c. Environmental: n/a.

3. Standard. The officer shall:

   a. explain SSBA rules and regulations IAW references to include:

      (1) diving responsibilities;
      (2) safety regulations; and
      (3) diving communications.

4. Time. 2 x 50-minute periods.

5. Method/Media.

   a. lecture - 1 period; and
   b. review / EC - 1 period.

6. References. A4.

7. Training Aids.

   a. Chalkboard;
   b. OHP; and
   c. Manual.

8. Test Details. Written test. Pass mark is 80 %

9. Remarks Nil

10. Teaching Points.

    a. Explain SSBA diving responsibilities IAW ref A4, Chap 1 to include:

       (1) CF diving organization;
       (2) Commanding Officer;
       (3) Diving Officer;
       (4) Diving Supervisor;
       (5) Diver;
       (6) Standby Diver; and
       (7) Diving Attendants.

b. Explain SSBA safety regulations IAW ref A4, Chap 1 to include:

(1) minimum personnel required;
(2) depth limitations;
(3) lifelines;
(4) umbilical breathing gas hoses;
(5) divers in boats underway;
(6) exceptional exposures;
(7) diving equipment maintenance;
(8) valves; and
(9) diving knife.

c. Explain diving communications IAW ref A4, Chap 1 to include:

(1) voice communications; and
(2) manual signals.

EC 406.1

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom.

2. OHP / PowerPoint.

ARRANGEMENTS AND PROCEDURES

3. The officer will complete a written test on SSBA rules and regulations IAW classroom lecture.

4. Assistance and references are denied.

5. Pass mark is 80%.

6. Time required - 1 period.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of SSBA rules and regulations.

2. Assistance and references are denied.

3. Pass mark is 80%.

4. Time required - 1 period.

PART III - ENABLING CHECK NOTES

1. Test Plan:

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE (%)
Diving Responsibilities	CF diving organization.	4
	Commanding Officer.	4
	Diving Officer.	15
	Diving Supervisor.	20
	Diver.	4
	Standby Diver	10
	Attendants	5
Safety Regulations	Personnel requirements.	6
	Depth limitations.	4
	Lifelines / umbilicals.	4
	Divers in boats underway.	3
	Exceptional exposures.	3
	Diving equipment.	5
Diving Communications	Voice communication / procedures.	8
	Manual signals.	5

EO 406 .2

1. Performance. Explain SSBA Equipment / SOPs.

2. Conditions.

   a. Given:

      (1) equipment; and
      (2) technical support.

   b. Denied:

      (1) assistance; and
      (2) references.

   c. Environmental: n/a.

3. Standard. The officer shall:

   a. describe SSBA equipment IAW references to include:

      (1) breathing apparatus;
      (2) diving suits and accessories;
      (3) underwater communications; and
      (4) support equipment; and

   b. explain SSBA SOPs.

4. Time. 10 x 50-minute periods.

5. Method/Media.

   a. lecture / demonstration - 6 periods;
   b. review - 2 periods; and c. EC - 2 periods.

6. References. A4, A17, A19, A20, A21.

7. Training Aids

   a. SL17B helmet complete including neck dam and liner;
   b. available dry suits (compressible and/or incompressible)
   c. hot water suit with boots, gloves, liners;
   d. Windax vest with weights, boots;
   e. Helle 3340 communications set; and
   f. whiteboard, OHP and manuals.

8. Test Details. Written test, pass mark is 80%.

9. Remarks. Not used.

10. Teaching Points.

a. Explain the SU 7B helmet to include:

(1) SSBA helmet;
(2) helmet weights;
(3) helmet interior;
(4) breathing system;
(5) regulator adjustment;
(6) emergency gas;
(7) helmet / oral-nasal fit;
(8) neck clamp, yoke and dam;
(9) helmet communications;
(10) hot water shroud; and
(11) donning the helmet.

b. Explain diving suits and accessories to include:

(1) types of suits;
(2) foam neoprene dry suit;
(3) incompressible dry suit;
(4) hot water suit;
(5) weighted vest (Windax);
(6) boots;
(7) swim fins; and
(8) underwater lights.

c. Explain underwater communications equipment IAW refs to include:

(1) Helle 3340 Set; and
(2) Helle 3342 (Helium unscrambler).

d. Explain SSBA support equipment IAW refs to include:

(1) Koh-I-nor hose;
(2) umbilical end tailing;
(3) hot water heater;
(4) diving stage;
(5) descending lines;
(6) dive ladder;
(7) air compression and storage;
(8) He02 / 02 storage;
(9) dive console; and
(10) pneumofathometer.

e. Explain dive procedures IAW references to include:

(1) attendant's duties;
(2) diving communications;
(3) descent by line;
(4) descent by stage;
(5) descent rate;
(6) gas supply pressure;
(7) on bottom;
(8) around corners, tunnelling or inside wrecks;

(9) under ships;

(10) lines and rigging;

(11) tools;

(12) preparing for ascent;

(13) ascent;

(14) decompression stops;

(15) current or tideway;

(16) bottom searches;

(17) cold weather operations;

(18) ice diving in SSBA; and

(19) hot water heater -use.

Note: refer to references to facilitate further clarification.

EC 406.2

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom.

2. OHP / PowerPoint.

ARRANGEMENTS AND PROCEDURES

3. The officer will complete a written test on SSBA equipment and Standard Operating Procedures IAW classroom lecture and instructor demonstrations.

4. Assistance and references are denied.

5. Pass mark is 80%.

6. Time required - 2 periods.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of SSBA equipment and Standard Operating Procedures.

2. Assistance and references are denied.

3. Pass mark is 80%.

4. Time required - 2 periods.

PART III - ENABLING CHECK NOTES

1. Test Plan:

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE (%)
Dive Procedures	Attendant's duties	4
	Descending;	4
	Gas supply pressure	4
	On bottom	4
	Around corners, tunnelling inside wrecks	4
	Tools	4
	Ascending	4
	Decompression stops	6
	Current or tideway	4
	Cold weather operations	4
	Ice diving	4
	Hot water heater	4
SSBA Breathing Apparatus	SL17B helmet components	10
	SL17B helmet procedures	8
Diving Suits and Accessories	Dry suits	3
	Hot water suits	3
	Windax weighted vest	3
	Diving accessories	3
SSBA Support Equipment	Umbilical hoses	5
	Stage procedures	5
	He02 / 02 / air compression and storage	5
	Dive console	10

PART IV - ENABLING CHECKLIST Not used.

PO 406 Supervise Surface Supplied Breathing Apparatus (SSBA) Operations.

EO 406 .3

1. Performance. Complete SSBA Air Chart work / Tables.

2. Conditions.

   a. Given:

      (1) references;
      (2) dive charts; and
      (3) scenario.

   b. Denied: assistance.

   c. Environmental: n/a.

3. Standard. The officer shall complete air chart work /tables IAW references to include an emergency situation and repetitive dives.

4. Time. 19 x 50-minute periods.

5. Method/Media.

   a. lecture / demonstration - 14 periods; and
   b. review / EC - 5 periods.

6. References. A4.

7. Training Aids. OHP and manuals.

8. Test Details. Written test, pass mark is 80 %

9. Remarks. This EO is taught by DCIEM instructor at FDU(A).

10. Teaching Points.

   a. Explain air dive charts to include;

      (1) air dive chart data to include:

         (a) times for:

            i. left surface;
            ii. reach bottom;
            iii. left bottom;
            iv. max time to 1st stop;
            v. delays;
            vi. reached 1st stop; and
            vii. reached surface;

         (b) total decompression time;
         (c) table and schedule used; and
         (d) total time of dive;

(2) maximum safe depth;

(3) computation of table;
(4) maximum oxygen;
(5) descent / ascent record;
(6) emergency air column;
(7) depth of stops;
(8) decompression time:

(a) in water; and
(b) chamber;

(9) time records:

(a) in water; and
(b) chamber; and

(10) rates of ascent in FPM:

(a) in water; and
(b) chamber.

b. Complete air dive charts to include:

(1) emergencies; and
(2) repetitive dives.

EC 406.3

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom.

2. OHP / PowerPoint.

ARRANGEMENTS AND PROCEDURES

3. The officer will complete a written test on information placed or found on an He02 dive chart IAW classroom lecture.

4. Assistance is denied; appropriate diving tables / charts are to be provided.

5. Pass mark is 80%.

6. Time required - 1 period.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of information placed or found on an He02 dive chart.

2. Assistance is denied; appropriate diving tables / charts are to be provided.

3. Pass mark is 80%.

4. Time required - 1 period.

PART III - ENABLING CHECK NOTES

1. Test Plan:

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE
He02 Chart Info	Chart data (i.e. name, etc.)	5
	Max safe depth	10
	Computation of table	10
	Max 0_2	10
	Effective atmospheres	10
	Ascent / descent rates - water / chamber	20
	Emergency air	10
	Emergency He O_2	
	Stops-water / chamber	15

PO 406 Supervise Surface Supplied Breathing Apparatus (SSBA) Operations.

EO 406 .4

1. Performance. Complete SSBA He02 Chart work / Tables.

2. Conditions.

   a. Given:

      (1) references;
      (2) dive charts; and
      (3) scenario.

   b. Denied: assistance.

   c. Environmental: n/a.

3. Standard. The officer shall complete He02 chart work / tables to include an emergency situation and repetitive dives.

4. Time. 2 x 50-minute periods.

5. Method/Media.

   a. lecture / demonstration - 1 period; and
   b. review / EC - 1 period.

6. References. A4.

7. Training Aids. OHP and manuals.

8. Test Details. Written exam. Pass mark is 80 %

9. Remarks. This EO is taught by DCIEM instructor at FDU(A).

10. Teaching

   a. Explain Heo2air dive charts including:

      (1) He02 dive chart data, to include:

         (a) time for:

            i. left surface;
            ii. reached bottom;
            iii. left bottom;
            iv. max time to 1st stop;
            v. delays;
            vi. reached 1st stop; and
            vii. reached surface;

         (b) total decompression time;

         (c) table and schedule used; and

         (d) total time of dive;

(2) maximum safe depth;
(3) computation of tables;
(4) maximum oxygen;
(5) descent / ascent record;
(6) emergency air column;
(7) depth of stops;
(8) decompression time to include:

(a) in water; and
(b) chamber;

(9) emergency He02 time records to include:

(a) in water; and
(b) chamber; and

(10) rates of ascent in FPM to include:

(a) in water; and
(b) chamber.

b. Complete He02 dive charts to include emergency situation and repetitive dives.

EC 406.4

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom.

2. Diving tables.

3. Blank charts.

4. OHP / Power Point.

ARRANGEMENTS AND PROCEDURES

5. The officer will complete a written test on information placed or found on an He02 dive chart IAW classroom lecture.

6. Assistance is denied; appropriate diving tables / charts are to be provided.

7. Pass mark is 80%.

8. Time required - 1 period.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your knowledge of information placed or found on an He02 dive chart.

2. Assistance is denied; appropriate diving tables / charts are to be provided.

3. Pass mark is 80%.

4. Time required - 1 period.

PART III - ENABLING CHECK NOTES

1. Test Plan:

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE
HeO2 Chart Info	Chart data (i.e. name, etc)	5
	Max safe depth	10
	Computation of table	10
	Max O2	10
	Effective atmospheres	10
	Ascent/ descent rates (water/RCC)	20
	Emergency air	10
	Emergency He02	10
	Stops - water / chamber	15

PART IV - CHECKLIST. N/A
PO 406 Supervise Surface Supplied Breathing Apparatus (SSBA) Operations.

EO 406.5

1. Performance. Explain SSBA Diving Emergencies and Responses.

2. Conditions.

   a. Given: nil.

   b. Denied: assistance and references.

   c. Environmental: n/a.

3. Standard. The officer shall explain diving emergencies/ responses IAW refs, including:

   a. general emergencies;
   b. equipment related emergency procedures; and
   c. emergency responses.

4. Time. 8 x 50-minute periods.

5. Method/Media.

   a. lecture - 6 period; and
   b. review / EC - 2 period.

6. References. A4.

7. Training Aids.

   a. Whiteboard;
   b. OHP; and
   c. manuals.

8. Test Details. Written test. Pass mark is 80 %

9. Remarks This EO is taught by DCIEM instructor at FDU(A).

10. Teaching Points.

   a. Explain diving emergencies in general, to include:

      (1) diving emergencies; and
      (2) respiratory emergencies.

   b. Explain equipment related emergency procedures to include:

      (1) introduction;
      (2) equipment failure during dive;
      (3) CO2 build-up;
      (4) excessive breathing resistance;
      (5) freeze-up;
      (6) flooded helmet / mask;
      (7) flooded suit;
      (8) lost communication;
      (9) blow up;
      (10) loss of main gas supply to include;

(a) loss of HeO2;
(b) loss of air;
(c) loss of O2 at 30 fsw in-water; and
(d) loss of O2 at 49 fsw in RCC; and

(11) fouling;

c. Explain emergency responses IAW the Aide-Memoire, Annex 2A, to include the following causes:

(1) traumatic injury in water;
(2) unconscious diver;
(3) drowning;
(4) over-exertion;
(5) O2 symptoms;
(6) fouling;
(7) lost communications;
(8) ditching and emergency ascent.

d. Explain omitted decompression in emergencies.

EC 406.5 Explain SSBA diving emergencies and responses

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILIITES REQUIRED

1. Classroom.

2. OHP/ Power Point

ARRANGEMENTS AND PROCEDURES

3. The officer will complete a written test on He02 diving emergency procedures IAW classroom lecture.

4. Assistance and references are denied.

5. Pass mark is 90 %

6. Time required - 2 periods.

PART II - INSTRUCTIONS GIVEN TO TRAINEE.

1. This EC will confirm your knowledge of He02 diving emergency procedures IAW classroom lecture.

2. Assistance and references are denied.

3. Pass mark is 90 %

4. Time required - 2 periods.

PART III - ENABLING CHECK NOTES

1. Test Plan:

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE (%)
Emergency procedures	Lost gases.	45
	02 toxicity.	45
	omitted decompression	10

PO 406 Supervise Surface Supplied Breathing Apparatus (SSBA) Operations.

EO 406.6

1. Performance. Conduct SSBA dives

2. Conditions.

   a. Given:

      (1). surface supplied diving support equipment/ systems;
      (2) dive team;
      (3) references/ SOPs
      (4) supervision; and
      (5) support platform.

   b. Denied: assistance with the exception of performance as tender.

   c. Environmental: in existing weather conditions, day and night.

3. Standard. The officer shall:

   a. conduct SSBA dives IAW references to include:

      (1) air dives; and
      (2) He02 dives.

   b. Perform the duties of;

      (1) console/ pneumo operator;
      (2) hot water heater operator;
      (3) crane/ winch operator;
      (4) communications;
      (5) tender;
      (6) diver; and
      (7) chart keeper.

   c. Perform equipment/ system pre/post-dive routines IAW applicable SOPs/ checklists.

   d. Perform each duty a minimum of twice.

   e. Dive to maximum operational depth of the limits of equipment in use IAW references.

4. Time. 28 x 50-minute periods.

5. Method/Media.

   a. lecture - 2 periods;
   b. demonstration - 5 periods; and
   b. review / EC - 1 period.

6. References. A4.

7. Training Aids. not used.

8. Test Details. This EO is tested as EC 406.6 (practical) and done in conjunction with EO 406.7

9. Remarks The instructor will require at least one QL6B to assist in monitoring trainees as EO 406.6 and 406.7 are done concurrently.

10. Teaching Points.

a. Explain and demonstrate SSBA IAW references to include:

(1) bringing air and gas systems online;
(2) console/ pneumo operations;
(3) stage and crane/ winch operations IAW SOPs and ref A4;
(4) tenders duties, to include;

(a) handling of umbilical; and
(b) donning suits and accessories;

(5) SL17B procedures to include;

(a) pre-dive checks;
(b) dressing;
(c) breathing system check; and
(d) helmet doffing.

(6) hot water heater operations, start up (NOTE: stress purging and restarting warning);

(7) communication check; and

(8) chart man.

b. Explain mixed gas standard procedures to include:

(1) helmet diver;
(2) systems check;
(3) shift to gas;
(4) dive;
(5) ascent;
(6) shift to air;
(7) 02 stop (NOTE - stress 02 caution);
(8) surfacing; and
(9) Sur-D.

c. Explain modification of mixed gas procedures for air dives IAW SOP Annex 5A.

d. Shutdown procedures for all SSBA and support equipment.

EC 406.6

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Surface supplied diving support vessel.

2. Support team.

3. Oxygen and He02

ARRANGEMENTS AND PROCEDURES

4. The officer will perform in all the positions for both air and mixed gas diving with the exception of supervisor.

5. This EO will be done concurrently with EO 406.7

6. Pass mark is SAT on checklist.

7. Assistance is denied with the exception of performing as tender.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your abilities to perform as a member of a SSBA dive team.

2. Pass mark is SAT on the checklist.

3. Assistance is denied with the exception of performing as tender.

PART III - ENABLING CHECK NOTES. Not used

PART IV - ENABLNG CHECKLIST.

EC 406.6

Student Date. SAT/UNSAT.

CHECKLIST	S	U	COMMENTS
COMMUNICATIONS			S=Sat, U=Unsat
1. Checks power supply (battery level)*			
2. Connects comms to umbilicals properly			
3. Does comm. check with all divers *			
4. Operates only on command from supervisor			
5. Switches to air/ He02 mode			
6. Carries out periodic checks with divers on travelling *			
TENDER			
1. Helps dress diver			
2. Ensures cleanliness of station			
3. Gauges cylinders *			
4. does proper reporting of diver			
5. Maintains control of lifeline (2 hands) *			
6. On Sur-D does proper undressing procedures			
7. Maintains control of diver on Sur-D *			
DIVER			
1. Checks out own diving gear *			
2. Does comm. check			
3. Employs proper lifeline signals			
4. Performs leak check on water entry *			
5. reports condition on diving (OK red)			
6. repeats all orders			
7. Carries out proper 02 procedures on 02 stop *			
8. Hooks and unhooks bales at proper times			
CONSOLE OPERATOR			
1. Ensure all banks on line (pressure recorded) *			
2. Identifies primary and secondary bank.			
3. Follows procedure for gas shift (if applicable) *			
4. Follows dive to bottom (over-bottom pressure) *			
5. Operates console with one hand			
6. Analyses gases (He02) *			
CRANE/ WINCH OPERATOR			
1. Starts crane			
2. Maintains rate of ascent and descent *			
3. Operates appropriate levers			
4. Operates only on command from supervisor *			

HOT WATER HEATER OPERATOR			
1. Fuel tanks full (kerosene)			
2. Plugs in unit (knows breaker location) *			
3. Starts pump (ensures water circ)			
4. Flashes up heater			
5. Sets temperature level			
6. Maintains heater operation *			
7. Ensures water level is maintained *			
CHART KEEPER			
1. Table used.			
2. Schedule used			
3. Dive depth			
4. Event time correctly recorded			
5. events correctly recorded on charts			

PO 406 Supervise Surface Supplied Breathing Apparatus (SSBA) Operations.

EO 406.6

1. Performance. Supervise SSBA dives

2. Conditions.

a. Given:

(1) dive scenario, to include an emergency situation;
(2) surface supplied diving equipment;
(3) dive platform;
(4) dive team;
(5) RCC equipment;
(6) surface support equipment;
(7) diving tables/references/ SOPs
(8) supervision for safety; and

b. Denied: assistance.

c. Environmental: day and night.

3. <u>Standard.</u> The officer shall supervise SSBA dives to include:

a. a minimum of five (5) SSBA air dives including:

(1) standard air decompression;
(2) in-water 02 decompression; and
(3) Sur-D;

b. a minimum of four (4) SSBA He02 dives including;

(1) in-water 02 decompression; and
(2) Sur-D.

c. formulate dive plans, including emergencies, for all environmental conditions;

(1) gas requirements;
(2) dive task;
(3) dive team;
(4) SSD equipment;
(5) decompression tables;
(6) SOPs; and
(7) emergency procedures.

d. brief dive team to include;

(1) dive plan; and
(2) responsibilities and duties of

(a) divers;
(b) tenders;
(c) hot water heater operator;
(d) gas panel operator;
(e) comms;
(f) winch operator;
(g) standby diver;

(h) RCC operator and 02 tender;
(i) chart/ timekeeper; and
(j) standby diver.

e. safely conducts a dive IAW a dive plan;

f. maintain control of dive team throughout the dive;

g. carry out emergency action;

h. carry out reports and return; and

i. complete CF incident/ accident reports.

4. Time. 70 x 50-minute periods.

5. Method/Media. practical

6. References. A4, A5, A9, A17, A19 - A22, A51, C6, C7.

7. Training Aids. not used.

8. Test Details. Pass mark is SAT on checklist

9. Remarks The instructor will require at least one QL6B to assist in monitoring trainees as EO 406.6 and 406.7 are done concurrently.

10. Teaching Points.

a. Explain duties and responsibilities of the diving supervisor including:

(1) maintaining control of the dive station; and
(2) monitoring environment and traffic.

b. Conduct demonstration dives, using trainee to fill positions including:

(1) standard air decompression;
(2) in-water decompression; and
(3) Sur-D.

EC 406.7

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Done currently with EO 406.6

ARRANGEMENTS AND PROCEDURES

2. The officer is to supervise SSBA IAW references.

3. This EO will be done concurrently with EO 406.6

4. The DCIEM instructor will assist the instructor with this EC.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will confirm your abilities to supervise SSBA diving using air and He02.

2. You are denied assistance.

3. Pass mark is SAT on checklist.

PART III - ENABLING CHECK NOTES.

An assistant instructor QL6b qualified will be required to monitor trainees during this EC.

PART IV - ENABLING CHECKLIST

DATE

NAME

CHECKLIST	S	U	COMMENTS
PRE-DIVE BRIEFING			
1. Type of dive (depth, BT etc...)			
2. Gas available and pressure			
3. Emerg procedures (lost gas,02 symp, etc..) *			
4. Standby diver briefed *			
PRE-DIVE CHECKS			
1. Conformed to SOP (gas shift) *			
2. On/ off emergency mode *			
3. System check - confirmed on He02/ air *			
DIVE PROCEDURES			
1 Ventilate divers (at depth) prior to leaving depth			
2. Depth and bottom time to charts *			
3. Inform dive site of the first stop-time to stop-rate travel			
4. Respond to emergency scenario *			
5. One minute warning to leaving bottom- stops			
6. Maintained rate of travel *			
7. Gave appropriate warning for shift to 02 console *			
8. Shift to 02 within 3 minutes *			
9. Informed dive site within “Emergency Sur-D” *			
10. Informed dive site within “Normal Sur-D” *			
11. Maintained control of the dive site.			

a. * Denotes a critical check
b. Trainee must score a SAT on all critical checks to achieve an overall grade of SAT
c. A score of UNSAT on 2 non-critical checks will be reviewed by the senior instructor/TRG CPO to determine the effect on the overall grade

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Surface supplied diving support vessel.

2. Support team.

3. Oxygen and He02

ARRANGEMENTS AND PROCEDURES

4. The officer will supervise SSBA diving operations IAW all applicable SOPs and references.

5. EC 406.6 and 406.7 must be successfully completed before beginning this PC.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This PC will confirm your abilities to supervise SSBA diving operations using He02 and air

2. Pass mark is SAT on the checklist.

3. Assistance is denied

PART III - ENABLING CHECK NOTES. The instructor remains overall responsible for the safety of the dive team.

PART IV - PERFORMANCE CHECKLIST

DATE

NAME

CHECKLIST	S	U	COMMENTS
SUPERVISE AIR DIVES			S=Sat, U=Unsat
1. Standard air decompression			
2. In-water 02 decompression			
3. Sur-D			
SUPERVISE He02 DIVES			
1. In-water decompression			
2. Sur-D			

PO 407 Supervise Battle Damage Repair (BDR) Operations.

EO 407.1

1. Performance. Weld Steel Using Electric-Arc Equipment.

2. Conditions.

   a. Given:

      (1) supervision for safety;
      (2) 1 /4" mild steel plate;
      (3) arc welding equipment;
      (4) grinder; and
      (5) protective equipment.

   b. Denied:

      (1) references; and
      (2) assistance.

   c. Environmental: n/a.

3. Standard. The officer shall:

   a. weld the following in the flat, vertical and overhead position:

      (1) 6" single vee butt joint (beads flat position);
      (2) 6" lap joint; and
      (3) 3/8 plate.

   b. prepare the pieces for the single vee butt joint so that the two edges to be welded have an angle of 45 degrees ::E have a root length of 1 /16" to 1 /8"; and

4. Time. 30 x 50-minute periods).

5. Method/Media.

   a. lecture / demonstration - 28 periods; and
   b. EC - 2 periods.

6. References. A57 - A60

7. Training Aids

   a. OHP; and
   b. equipment IAW condition statement

8. Test Details**.** Written exam pass mark is 80%. The item must meet visual inspection specification.

9. Remarks. This EO is conducted at CFNES (H) welding shop for a total of 5 training days. This phase is not to produce a professional welder but to make him / her aware of the difficulties encountered with underwater welding and ensure he /she possesses all safety knowledge required to supervise underwater welding.

10. Teaching Points.

a. Explain the safety precautions to be observed while carrying out arc welding to include:

(1) hazards;
(2) work area;
(3) equipment;
(4) ventilation;
(5) protective clothing and equipment;
(6) certifying heat affected compartments;
(7) fire sentries; and
(8) fire inspection - post operation.

b. Explain the theory of electric arc welding to include:

(1) principles of arc welding;
(2) power source / supply;
(3) polarity;
(4) categories of welders;
(5) consideration for selection of electrodes;
(6) function of slag;
(7) rod coatings; and
(8) rod identification.

c. Explain electrode manipulation to include:

(1) starving and maintaining an arc; and
(2) weaving method;

d. Explain the causes and cures of common welding problems to include:

(1) porous welds;
(2) cracked welds;
(3) undercutting;
(4) distortion;
(5) spatter;
(6) lack of fusion;
(7) overlapping;
(8) poor penetration; and
(9) magnetic arc blow.

e. Explain the following:

(1) common terms applied to welds;
(2) types of welds;
(3) types of joints; and
(4) welding positions.

f. Explain the equipment and safety preparations to be carried out prior to striking an arc.

g. Explain non-destructive testing.

EC 407.1

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Material and facilities supplied by CFNES (Hfx) welding shop.

ARRANGEMENTS AND PROCEDURES

2. The officer will weld steel. The instructor will provide supervision for safety.

3. References and assistance are denied.

4. Pass mark is a grading of SAT on EC 407 .1 checklist & 80% on the written exam.

5. Time required - 5 training days.

6. This EO is conducted at CFNES (Hfx) welding shop.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will demonstrate your capability to weld steel.

2. References and assistance are denied.

3. Pass mark is a grading of SAT on EC 407 .1 checklist.

4. Time required - 5 training days.

PART III - ENABLING CHECK NOTES

1. Not used.

PART IV - ENABLING CHECK LIST

Weld Steel.

Student: Date: SAT/UNSAT:

CHECKLIST	S	U	COMMENTS
1. Horizontal Position			
a. Run stringer bead			
b. Weld lapp joint			
c. Weld tee joint			

Notes:

a. One UNSAT in the welding phase constitutes an overall grading of UNSAT.

b. Welds must meet visual inspection specifications.

PO 407 Supervise Battle Damage Repair (BDR) Operations.

EO 407.2

1. Performance. Explain Underwater Tool Safety and Operation.

2. Conditions.

   a. Given:

      (1) diving equipment and team;
      (2) underwater tools and equipment;
      (3) technical support;
      (4) SOPs; and
      (5) supervision.

   b. Denied: assistance and references.

   c. Environmental: n/a.

3. Standard. The officer shall explain SOPs for underwater tools to include:

   a. a/w welding equipment;
   b. KERI cable;
   c. hydraulic tools to include:

      (1) cutting;
      (2) drilling;
      (3) tapping; and
      (4) grinding;

4. Time. 44 x 50-minute periods.

5. Method/Media.

   a. lecture - 18 periods;
   b. practical - 24 periods; and
   c. EC - 2 periods.

6. References A3, A4, A18, A46, A49, A54.

7. Training Aids.

   a. whiteboard;
   b. OHP;
   c. PowerPoint;
   d. hydraulic tools and equipment; and
   e. underwater welding equipment.

8. Test Details. Written test. Pass mark is 80%.

9. Remarks. The practical phase of this EO may be done in conjunction with PO 406.

10. Teachings Points.

    a. Stress the importance of ensuring that all safety precautions for underwater cutting and welding are taken, and being on guard for hazards, including those not specifically covered.

b. Explain the general cutting and welding precautions to include:

(1) qualified personnel;
(2) approved equipment;
(3) an attendant assigned to the care and control of the safety switch and butting gas supplies;
(4) ensuring the work area is free of highly combustible or explosive materials;
(5) ensuring the proper use of oxygen/oxygen handling equipment; and
(6) ensuring no other work interferes with or endangers the diver carrying out the cutting or welding;

c. explain the mandatory use of a positive operating disconnect safety switch;
d. explain the precautions related to the use of power cables and cable connectors for underwater use;
e. explain the diving dress and equipment with regard to protecting the diver against electrical shock and eye injury;
f. explain the underwater procedures when cutting and welding;
g. explain the precautions to be taken against explosive gases; and
h. explain additional hazards the diver may encounter.

Hydraulics:

a. review the following:

(1) the principle on which hydraulics is based - Pascal's Law;
(2) the relationship of force, pressure and area; and
(3) the fundamental law of physics that states energy can neither be destroyed nor created and the relationship between force, distance and speed;

b. briefly explain hydraulic power transmission to include:

(1) the definition of hydraulics;
(2) the input and output of a system; and
(3) source of power and prime mover;

c. explain some advantages of hydraulics to include:

(1) variable speed;
(2) reversible;
(3) overload protection;
(4) small size;
(5) can be stalled;
(6) virtually unaffected by depth; and
(7) no bubbles and noise level is insignificant;

d. explain the primary purposes of hydraulic fluid:

(1) to transmit power;
(2) to lubricate moving parts;
(3) to seal clearances between moving parts; and
(4) to cool off and dissipate heat;

e. use hydraulic circuit drawings to explain:

(1) how speed is varied; how drives are reversed;
(2) how relief valves work to protect a system; and
(3) how directional valves work;

f. briefly explain what components comprise the Model 1 Diesel hydraulic power unit to include:

(1) an **Onan Model DJ120** diesel engine;
(2) a Denison Model TIC-011-2112 pump;
(3) a 12-volt commercial battery;
(4) a skid mounted frame assembly;
(5) associated hydraulic equipment, including valves, gauges, fittings, hoses, 30-gallon hydraulic oil reservoir, filters; and
(6) electrical engine controls;

g. briefly explain the hydraulic circuit, to include:

(1) diesel engine;
(2) hydraulic pump;
(3) flow regulator;
(4) relief valves;
(5) hydraulic flow meter;
(6) pressure gauge;
(7) oil filter; and
(8) directional control valves;

h. explain the following procedures for operating the diesel hydraulic power unit:

(1) starting;
(2) stopping;
(3) high and low temperatures;
(4) setting for tools with a maximum pressure of 2000 psi;
(5) setting for tools with a maximum pressure less than **2000** psi;
(6) setting flow rate;
(7) changing tools; and
(8) emergency turn-off; and

i. briefly explain scheduled maintenance is based on hourly usage, as outlined in the operator's manual to include:

(1) maintenance is broken up onto two sections:

(a) diesel engine; and
(b) hydraulic system; and

(2) the maintenance procedure is divided into two categories:

(a) operator maintenance - performed by the operator; and
(b) critical maintenance -performed by qualified personnel.

Equipment and procedures:

a. explain the equipment requirements for underwater shielded arc welding are basically the same for oxygen / arc cutting, to include:

(1) same power requirements; and
(2) why DC power is recommended over AC power;

b. explain torches and electrode holders currently used in the Navy for underwater welding:

(1) Arcair Sea Stinger Underwater Welding Electrode Holder; and
(2) Arcair Underwater Sea Torch Combination Torch and Arcair Underwater Torch may be used for welding when fitted with appropriate collets;

c. explain the sea weld electrode currently in use to include:

(1) the purposes of flux coating; and
(2) waterproof coating;

d. explain adverse welding conditions to include:

(1) dirty surfaces;
(2) multiple pass welding;
(3) unstable platform;
(4) poor fit; and
(5) low visibility;

e. explain the self-consuming or drag technique to produce fillet welds to include:

(1) the relationship between leg size and electrode diameter; and
(2) the advantage of fillet welds;

f. explain the drag welding procedure;

g. explain the procedures for overhead and vertical position welding, using the drag technique;

h. explain the procedure for repairing small cracks; review safety precautions to be observed;

i. explain the importance of ensuring that the equipment is assembled as indicated, as warnings in the respective references, in order to avoid safety hazards and equipment damage;

j. explain the technique for piercing holes in steel plate; and

k. explain the techniques for cutting plate of different thicknesses;

1. explain the cutting of cast iron and non-ferrous metals; and

m. explain flexible thermal arc cable (KERI cable) to include:

(1) the process;
(2) equipment;
(3) application; and
(4) operation.

EC 407 .2

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Classroom.
2. Diving support vessel.
3. Hydraulic tools and associated equipment.
4. Support personnel.
5. Underwater welding equipment.

ARRANGEMENTS AND PROCEDURES

6. The officer will complete a written exam on: underwater welding /cutting safety pro-cedures; hydraulic equipment safety, operation and usage.

7. Pass mark for exam is 80%.

8. Assistance and references are denied.

9. The practical phase is not assessed. This phase is used to give the trainee on appreciation of the hazards and capabilities of underwater welding, cutting and hydraulic equipment.

10. This EO should be completed prior to commencing PO 406

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will test your knowledge of underwater welding / cutting safety procedures and hydraulic equipment safety operation and usage.

2. Assistance is denied.

3. Pass mark is 80%.

4. Time required - 2 periods.

PART III - ENABLING CHECK NOTES

1. Test Plan:

MAJOR CONTENT AREA	SPECIFIC CONTENT AREA	VALUE
WELDING / CUTTING	Safety	30
	Uses	5
	Procedures (SOPs)	15
HYDRAULICS	General	5
	Safety	30
	Circuit / Components	10
	Uses	5

PO 407 Supervise Battle Damage Repair (BDR) Operations.

EO 407.3

1. Performance. Use Underwater Video and Photography Equipment.

2. Conditions.

   a. Given:

      (1) photography equipment;
      (2) references;
      (3) technical support; and
      (4) assistance.

   b. Denied: n/a.

   c. Environmental: within the environmental limits contained in reference A3.

3. Standard. The officer shall set up the system / equipment, ensuring connections of all components are made IAW set up procedures to include:

   a. Nikonos equipment; and
   b. Sony Dive Buddy (Handy Cam).

4. Time. 7 x 50-minute periods.

5. Method/Media. lecture / demonstration.

6. References. A3, A50.

7. Training Aids.

   a. OHPs;
   b. slide projector; and
   c. one camera and flash kit / student.

8. Test Details. There is no specific EC for this EO.

9. Remarks. Request services of formation photographers. b. This is a basic level familiarization topic only.

10. Teaching Points.

    a. Give a brief description of the underwater video and photography equipment to include:

       (1) what the system is;
       (2) application;
       (3) power requirements; and
       (4) transportation requirements:

          (a) weight; and
          (b) space.

b. Explain the setup of the system, identifying each of the components:

(1) lamp;
(2) a/w video camera;
(3) control unit;
(4) assembly case;
(5) lights;
(6) camera specifications and characteristics;
(7) camera nomenclature;
(8) camera controls;
(9) camera preparation;
(10) camera operation;
(11) post use and diving checks;
(12) electronic flash specification;
(13) electronic flash nomenclature;
(14) flash controls;
(15) flash preparation;
(16) flash operation and camera hook-up;
(17) post usage and diving checks; and
(18) photography pointers and tips.

PO 407 Supervise Battle Damage Repair (BDR) Operations.

EO 407.4

1. Performance. Describe Underwater Search Equipment.

2. Conditions.

   a. Given:

      (1) underwater search equipment (ROV / Single Side Scan Sonar (SSS) / DSI; (2) references; and (3) technical support.

   b. Denied: n/a.
   c. Environmental: n/a.

3. Standard. The officer shall describe underwater search equipment to include:

   a. ROVs to include:

      (1) set up the ROV for operation;
      (2) carry out pre- and post-dive routines; and
      (3) DSI concept;
      (4) application;
      (5) system components; and
      (6) operating procedures.

   b. SSS to include:

      (1) side scan concept;
      (2) application;
      (3) system components;
      (4) operating / towing procedures; and
      (5) interpretation of side scan records; and

   c. DSI to include:

      (1) DSI concept;
      (2) application;
      (3) system components; and
      (4) operating procedures.

4. Time. 19 x 50-minute periods.

5. Method/Media.

   a. lecture / demonstration - 6 periods; and
   b. Performance - 13 periods.

6. References. A4, A11.

7. Training Aids.

   a. OHPs;
   b. handouts;
   c. thruster and components; and
   d. joystick.

8. Test Details. No specific EC for this EO. Handout given to facilitate homework assignment.

9. Remarks. This is a basic-level knowledge familiarization topic only. No EC required.

10. Teaching Points.

a. Give a brief description of underwater search equipment including:

(1) ROV:

(a) vehicle effectiveness;
(b) system reliability;
(c) specifications;
(d) applications; and
(e) transportation;

(2) DSI:

(a) vehicle effectiveness;
(b) system reliability;
(c) specifications;
(d) applications; and
(e) transportation; and

(3) SSS:

(a) vehicle effectiveness;
(b) system reliability;
(c) specifications;
(d) applications; and
(e) transportation.

b. Explain underwater search equipment operations to include:

(1) ROV:

(a) set up;
(b) deck space;
(c) vehicle operation;
(d) sonar operation
(e) pre- and post-dive inspections;
(f) launch and recovery;
(g) trim;
(h) associated equipment; and
(i) vehicle control;

(2) DSI:

(a) set up;
(b) deck space;
(c) vehicle operation;
(d) sonar operation
(e) pre- and post-dive inspections;
(f) launch and recovery;

(g) trim;

(h) associated equipment; and

(i) vehicle control; and

3. SSS

(a) set up;

(b) deck space;

(c) vehicle operation;

(d) sonar operation

(e) pre- and post-dive inspections;

(f) launch and recovery;

(g) trim;

(h) associated equipment; and

(i) vehicle control.

EC 407.4

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. ROV components.

2. Handout on ROV 555 and D51.

ARRANGEMENTS AND PROCEDURES

3. This EO will be tested in PC 407 .

4. The EC is not for pass / fail assessment

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. As per PC 407

PART III - ENABLING CHECK NOTES

1. Not used.

PO 407 Supervise Battle Damage Repair (BDR) Operations.

EO 407.5

1. Performance. Conduct Underwater Recoveries.

2. Conditions.

a. Given:

(1) recovery equipment, to include:

(a) all available lifting bags;
(b) all available lifting devices; and
(c) support equipment;

(2) dive team;
(3) recovery object / tasking;
(4) supervision;
(5) references; and
(6) technical support.

b. Denied: n/a.

c. Environmental: within environmental limits contained in A3.

3. Standard. The officer shall as a member of a team, carry out a minimum of two (2) underwater recoveries IAW equipment SOPs, to include:

a. a minimum of one (1) recovery as diver; and
b. a minimum of one (1) recovery as surface support.

4. Time. 4 x 50-minute periods.

5. Method/Media.

a. lecture/demonstration/performance - 1 period; and
b. practical - 3 periods.

6. References. A3, 133.

7. Training Aids.

a. whiteboard;
b. OHP; c. manuals; and d. available lifting bags.

8. Test Details. Practical assessment.

9. Remarks. This EO is to be conducted concurrently with EO 404.2 .

10. Teaching Points.

a. Explain lifting tools.
b. Explain project considerations.
c. Obtain / explain available lifting bags.

d. Explain the capabilities and limitations of other recovery methods, to include:

(1) YDTs lifting booms and capstans;
(2) FDU training barges;
(3) dockyard floating crane; and
(4) Mark 2 Mod 1 Flotation Bladder.

e. Review required physics for recovery.

EC 407.5

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Diving support vessel.

2. Support personnel.

ARRANGEMENTS AND PROCEDURES

3. The officer will carry out underwater recoveries IAW A3 and B3.

4. Assistance and references are allowed.

5. Pass mark is a grading of SAT on the checklist.

6. Time required - 3 period.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will enable you to carry out underwater searches and recoveries safely.

2. You are allowed assistance and references.

3. Pass mark is a grading of SAT on the checklist.

4. Time allowed - 3 periods.

PART III - ENABLING CHECK NOTES

1. Not used.

PART IV - ENABLING CHECK LIST

EC 407 .5

Student: Date: _SAT/UNSAT

CHECKLIST	S	U	COMMENTS
1. Ensures all lifting devices are free of damage			
2. As a member of a team:			
a. Recovers item as diver			
b. Recovers item as surface support			
3. Observes all safety precautions			
4. Carries out post-dive routine of all lifting gear*			

Notes:

a. * Denotes critical check.

b. Trainee must score a SAT on all critical checks to achieve an overall grading of SAT.

c. A scoring of UNSAT on two non-critical checks will be reviewed by the Senior Instructor / Trg CPO to determine the effect on the overall grade.

PO 407 Supervise Battle Damage Repair (BDR) Operations.

EO 407 .6

1. Performance. Supervise Underwater Recoveries.

2. Conditions.

a. Given:

(1) recovery equipment, to include:

(a) all available lifting bags;
(b) all available lifting devices; and
(c) support equipment;

(2) dive team;
(3) recovery object / tasking;
(4) references; and
(5) SOPS.

b. Denied: assistance.

c. Environmental: within environmental limits contained in ref A3.

3. Standard. The officer shall supervise a minimum of one (1) underwater recovery IAW SOPs to include:

a. recovery plan;
b. team briefing;
c. plan execution; and
d. safety procedures.

4. Time. 3 x 50-minute periods.

5. Method/Media. performance.

6. References. A3.

7. Training Aids. Not used.

8. Test Details. Practical assessment.

9. Remarks. This EO is conducted concurrently with EO 404 .3 .

10. Teaching Points.

a. There is no additional knowledge required to support this EO.

EC 407.6

PART I - ADMINISTRATIVE INSTRUCTIONS

MATERIAL AND FACILITIES REQUIRED

1. Diving support vessel.

2. Support personnel.

ARRANGEMENTS AND PROCEDURES

3. The officer will supervise underwater recoveries IAW A3 and B3. Instructor will provide supervision for safety.

4. Assistance is denied.

5. Pass mark is a grading of SAT on the checklist.

6. Time required - 3 periods.

PART II - INSTRUCTIONS GIVEN TO TRAINEE

1. This EC will demonstrate your capability to supervise underwater recoveries.

2. You are denied assistance and references.

3. Pass mark is a grading of SAT on the checklist.

4. Time allowed - 3 periods.

PART III - ENABLING CHECK NOTES

1. Not used.

PART IV - ENABLING CHECK LIST

EO 407.6

Student: Date: SAT/UNSAT:

CHECKLIST	S	U	COMMENTS
1. Ensures all support equipment and lifting devices present.			
2. As supervisor:*			
a. Makes plan for recovery			
b. Briefs team			
c. Executes recovery IAW plan			
3. Adheres to all safety precautions *			

Notes:

a. * Denotes critical check.

b. Trainee must score a SAT on all critical checks to achieve an overall grading of SAT.

c. A scoring of UNSAT on two non-critical checks will be reviewed by the Senior Instructor / Trg CPO to determine the effect on the overall grade.

PO 408 Coordinate Mine Counter Measures (MCM) Operations.

EO 408.1

1. Performance. Related EOs taught through Eguermine, Ostende, Belgium.

PO 409 Supervise Explosive Ordnance Disposal (EOD) Operations.

EO 409.1

1. Performance. Conduct Render Safe Procedures (RSP) of Ground Ordnance.

2. Conditions.

   a. Given:

      (1) references;
      (2) EOD team;
      (3) EOD equipment; and
      (4) scenario / training ordnance.

   b. Denied: assistance.

   c. Environmental: within environmental limits authorized in references.

3. Standard.

   a. The officer shall conduct RSPs of ground ordnance to include:

      (1) grenades including:

         (a) identify grenades;
         (b) determine correct RSP for grenades; and
         (c) conduct correct RSP for grenades;

      (2) landmines including:

         (a) identify landmines;
         (b) determine correct RSP for landmines; and
         (c) conduct correct RSP for landmines;

      (3) rockets including:

         (a) identify rockets;
         (b) determine correct RSP for rockets; and
         (c) conduct correct RSP for rockets; and

      (4) projectiles including:

         (a) identify projectiles;
         (b) determine correct RSP for projectiles; and
         (c) conduct correct RSP for projectiles.

All further information is contained within the appropriate training establishments' documentation.

PO 409 Supervise Explosive Ordnance Disposal (EOD) Operations.

EO 409 .2

1. Performance. Conduct Render Safe Procedures (RSP) of Air Ordnance.

2. Conditions.

   a. Given:

      (1) references;
      (2) EOD team;
      (3) EOD equipment; and
      (4) scenario / training ordnance.

   b. Denied: assistance.

   c. Environmental: within environmental limits authorized in references.

3. <u>Standard.</u>

   a. The officer shall conduct RSPs of air ordnance to include:

- (1) aircraft explosive hazards (AEH) including:

      (a) identify AEHs;
      (b) determine correct RSP for AEHs; and
      (c) conduct correct RSP for AEHs;

   (2) bombs including:

      (a) identify bombs;
      (b) determine correct RSP for bombs; and
      (c) conduct correct RSP for bombs;

   (3) dispensers and payloads (D & Ps) including:

      (a) identify D & Ps;
      (b) determine correct RSP for D & Ps; and
      (c) conduct correct RSP for D & Ps; and

   4) missiles including:

      (a) identify missiles;
      (b) determine correct RSP for missiles; and
      (c) conduct correct RSP for missiles.

   All further information is contained within the appropriate training establishments documentation.

PO 409 Supervise Explosive Ordnance Disposal (EOD) Operations.

EO 409.3

1. Performance. Conduct Render Safe Procedures (RSP) of Underwater Ordnance.

2. Conditions

   a. Given:

      (1) references;
      (2) EOD team;
      (3) EOD equipment; and
      (4) scenario / training ordnance.

   b. Denied: assistance.

   c. Environmental: within environmental limits authorized in references.

3. Standard.

   a. The officer shall conduct RSPs of underwater ordnance to include:

      (1) contact mines including:

         (a) identify contact mines;
         (b) determine correct RSP for contact mines; and
         (c) conduct correct RSP for contact mines;

      (2) influence mines including:

         (a) identify influence mines;
         (b) determine correct RSP for influence mines; and
         (c) conduct correct RSP for influence mines; and

      (3) torpedoes including:

         (a) identify torpedoes;
         (b) determine correct RSP for torpedoes; and
         (c) conduct correct RSP for torpedoes.

All further information is contained within the appropriate training establishments' documentation.

ANNEX A - Suggested Sequencing and Timings

EO	# of Periods	# of Periods for EC	# of Periods for PC
401 .1	3	1	PC is Passed When all EOs are Successfully Completed
.2	9	1	
.3	30	As Per Instructor Profile	
.4	28	As Per Instructor Profile	
.5	24	As Per Instructor Profile	
403 .1			As Per CFSEME Lesson Plans
402 .1	13	3	As Per DCIEM Lesson Plans
.2	28	3	
.3	9	1	
.4	11	1	
.5	35	16	
404 .1	~2	1 ~	PC is Completed When PC Checklist is Complete
.2	19	4	
.3	10	3	
407 .3	7	Not Assessed	
.4	19	Not Assessed	
.5	4	3	
.6	3	3	
.1	30	As Per CFNES(H) Lesson Plan	
406 .1	2	1	PC is Completed When PC Checklist is Complete
.2	10	2	
.3	19	I 1	
.4	2	1	
.5	8	2	
.6	28	Done with 406 .7	
407 .2	44	2	
406 .7	70		
404 .4	8	2	PC is Complete When EOs are Successfully Completed.
.5	130	35	
.6	10		
.7	2	1	
.8	6	1	
.9	11	9	
.10	86	4	
.11	10	10	
405 .1	3	1	PC is Complete When EOs are Successfully Completed.
.2	5	1	
.3	TBP		
.4	TBP		

.5	14	14	
408	175		As Per TE Lesson Plans
409	511 -L		As Per TE Lesson Plans

ANNEX B - Course History

This annex is to record course history issues such as explanation of amendments, pertinent findings, trends, resource problems, and modifications. Focus should be given to both excellent and questionable issues of the course. Encapsulating important course history information is a useful managerial tool for amendment considerations by future course developers.

Document Approval Date Comments

Current document Summary of QSP board record of proceedings of 19 Dec 97 as follows:

1. OS recommendations: The OS is approximately four years old. Much has changed and safety is a key consideration, so many task, skill and knowledge items required amending. Most significant of these is the documentation of the CLDO supervision function.

An occupation analysis is required for MARS 71 D - CLDO, however, it should not commence prior to Summer '99 in order to assess the effects of FOND MIP and planned OST changes.

2. Related Training Recommendations: CLDOs have out-service training (OST) requirements which are critical to become CLDO qualified. For this reason, the QSPWB decided that these critical OST courses are Performance Objectives:

a. PO 409-Supervise EOD Operations. This is a 15-week OST course in Indianhead, MD which will soon re-locate to Eglin AFB, Florida; and

b. PO 408 - Coordinate MCMV Operations. This is a five-week course is Ostende, Belgium (CO MCMV). Currently, CLDOs receive similar training in the UK of the LMCDO course. The LMCDO course is attended via the SPQR route, however, after two years it is considered that this training (UK) is too specific to the RN HUNT class and it does not meet the requirements needed by CLDOs. The QSPWB recommends to incorporate the CO MCMV course in Belgium as it fulfils CLDO MCM training requirements and the course is shorter and less expensive than the current LMCDO course. The SPQR for LMCDO will be deleted upon approval of the CLDO QSP which incorporates the CO MCMV course.

3. Related In-service Training Recommendations: For the same reasons stated above in para 2.

a. PO 402 - Supervise Emergency Treatment of Diving Ailments. This is currently a four-week course at DCIEM, Toronto. All He02 related training currently instructed at DCIEM will be taught at FDU(A). The new DCIEM phase will be three weeks which will save $10K in TD costs; and

b. PO 403 - Conduct Basic EOD. This is a six-week course at CFSEME, CF Borden.

4. OS Train / No Train Decisions:

a. OS tasks requiring training: CT001 - CT007, CT009, CT012 - CT014, CS001, CS002, CS004, CS005, CS007, CS008, CS011, CS014, CS015, AK063, CK001 - CK005, CK008 - CK011, CK013 -CK017, CK021, CK024, CK025, CK027, CK028, CK029 and CK031. All training decisions were made by comparing the skills and knowledge of a Ship's Team /Diving Officers who commence CLDO training to levels stated or newly recommended in the OS;

b. OS tasks not requiring training: CT008, CT01 0, CT01 1, CT015, CS003, CS006, CS009, CS010, CS012, CS013, CK012, CK018, CK019, CK020, CK022, CK026, CK030. The rationale process for non-training was identical to that of the training decisions; and

c. Others: CL007 - Unsure of the statement (see para 1 b(11); and CK023 - Deleted (see para 1 b(14).

5. Training Strategy Decisions: The following items will be covered by on-job experience gained after CLDO course (i.e. CLDO qual is IAW CFITES methodology to train to the minimum standard);

a. Versatile Exercise Mine System (VEMS), a newly acquired system used to monitor magnetic and acoustic signatures;

b. ship repair;

c. ice diving is not specifically trained as it is covered in rules and regulations and experience will be gained during exercises; and

d. tender command.

6. Lessons Learned: QSP Board should be a minimum of two weeks; three optimum.

7. Other Considerations:

a. CLDO spend 18 months on course without a PER. They are at a disadvantage in comparison to their MARS peers for TE consideration.

b. FDU(A) and FDU(P) should hold copies of all In/Out-Service Training documentation. Also, those organizations should be formally informed to include FDUs on dist lists for future amendments.

c. Supervision was the focus for most POs of this course. The conduct in some cases is necessary but less important WIRT the skill level required, i.e. Supervision to level 2 and conduct to level 1.